NEET

Time : 3 Hrs
MM : 720

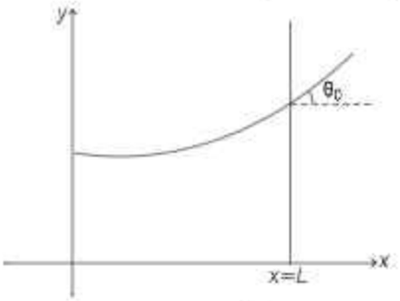
Solved Paper 2025

Instructions

- This question paper contains 180 Multiple Choice Question from the subjects **Physics, Chemistry & Biology.**
- There are 45 Questions from Physics section, 45 Questions from Chemistry section and 90 Questions from Biology section.
- Each question carries 4 **Marks**. For each correct response, the candidate will get 4 **Marks.** For each incorrect response **1 Mark** will be deducted from the total scores.

PHYSICS

1. Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z-direction. When filled with a liquid of surface tension S and density ρ, the liquid surface makes angle $\theta_0 (\theta_0 << 1)$ with the X-axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is



[take $\theta(x) = \sin\theta(x) = \tan\theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity]

(a) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}x$ (b) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}y$

(c) $\frac{d^2y}{dx^2} = \sqrt{\frac{\rho g}{S}}$ (d) $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}}x$

2. A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

(a) 100 (b) 125
(c) 150 (d) 250

3. An electron (mass 9×10^{-31}kg and charge 1.6×10^{-19} C) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field **B** of magnitude 9×10^{-4}T perpendicular to its direction of motion. We wish to apply an uniform electric **E** together with the magnetic field, so that the electron does not deflect from its path. Then (speed of light $c = 3 \times 10^8$ ms^{-1})

(a) **E** is perpendicular to **B** and its magnitude is 27×10^4 V m^{-1}.

(b) **E** is perpendicular to **B** and its magnitude is 27×10^2V m^{-1}.

(c) **E** is parallel to **B** and its magnitude is 27×10^2 V m^{-1}.

(d) **E** is parallel to **B** and its magnitude is 27×10^4 V m^{-1}.

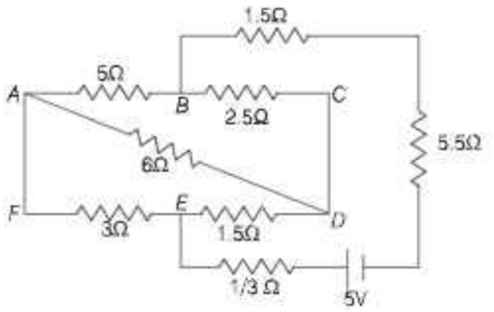
4. There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

(a) 0.25 (b) 0.40
(c) 0.5 (d) 0.75

5. The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying breaks, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the breaks on cars A and B, respectively, then the ratio F_A / F_B is

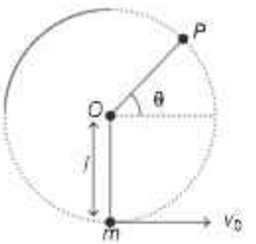
(a) 3/2 (b) 2/3
(c) 1/3 (d) 1/2

6. The current passing through the battery in the given circuit is



(a) 2.0 A (b) 0.5 A (c) 2.5 A (d) 1.5 A

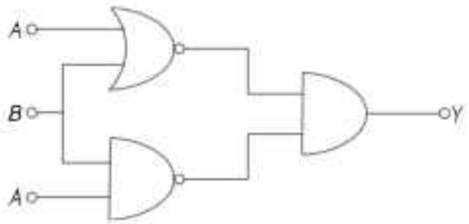
7. A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is



(a) $(\sin\theta)^{1/2}$ (b) $\left(\frac{1}{2+3\sin\theta}\right)^{1/2}$

(c) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{1/2}$ (d) $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{1/2}$

8. The output (Y) of the given logic implementation is similar to the output of an/a .....gate.



(a) AND (b) NAND (c) OR (d) NOR

9. The electric field in a plane electromagnetic wave is given by $E_z = 60\cos(5x + 1.5\times10^9\, t)$ V / m

Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field)

(a) $B_y = 2\times10^{-7}\cos(5x + 1.5\times10^9\, t)$ T

(b) $B_x = 2\times10^{-7}\cos(5x + 1.5\times10^9 t)$ T

(c) $B_z = 60\cos(5x + 1.5\times10^9 t)$ T

(d) $B_y = 60\sin(5x + 1.5\times10^9 t)$ T

10. A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is

(Take $g = 9.8$ m / s^2)

(a) 21 Ns (b) 7 Ns

(c) 0 (d) 84 Ns

11. AB is a part of an electrical circuit (see figure). The potential difference "$V_A - V_B$" at the instant when current $i = 2$ A and is increasing at a rate of 1 amp/s is



(a) 5 V (b) 6 V

(c) 9 V (d) 10 V

12. A 2A current is flowing through two different small circular copper coils having radii ratio 1:2. The ratio of their respective magnetic moments will be

(a) 1:4 (b) 1:2

(c) 2:1 (d) 4:1

13. In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively

(a) $4p$ and $4m$ (b) p^4 and $4m$

(c) $4p$ and m^4 (d) p^4 and m^4

14. An oxygen cylinder of volume 30 L has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27C°. The mass of the oxygen withdrawn from the cylinder is nearly equal to

[Given, $R = \frac{100}{12}$ J mol^{-1} K^{-1} and molecular mass of $O_2 = 32$, 1 atm pressure $= 1.01\times10^5$ N / m^2]

(a) 0.125 kg (b) 0.144 kg

(c) 0.116 kg (d) 0.156 kg

15. In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

(a) $-\frac{2}{(x+2)^3}$ (b) $-\frac{2}{(2x+1)^3}$

(c) $+\frac{2}{(x+1)^3}$ (d) $+\frac{2}{2x+1}$

16. To an ac power supply of 220 V at 50 Hz, a resistor of 20Ω, a capacitor of reactance 25 Ω and an inductor of reactance 45Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively

(a) 7.8 A and 30°

(b) 7.8 A and 45°

(c) 15.6 A and 30°

(d) 15.6 A and 45°

17. The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

(a) 100 days (b) 105 days
(c) 115 days (d) 108 days

18. A model for quantised motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

(a) $\frac{he}{\pi m}$ (b) $\frac{he}{2\pi m}$ (c) $\frac{heB}{\pi m}$ (d) $\frac{heB}{2\pi m}$

19. Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K. The left end of the combination is maintained at temperature $3T$ and the right end at T. The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2. The ratio T_1/T_2 is



(a) $\frac{3}{2}$ (b) $\frac{4}{3}$ (c) $\frac{5}{3}$ (d) $\frac{5}{4}$

20. The plates of a parallel plate capacitor are separated by d. Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates. If $K_1 = 1.25K_2$, the value of K_1 is

(a) 2.66 (b) 2.33 (c) 1.60 (d) 1.33

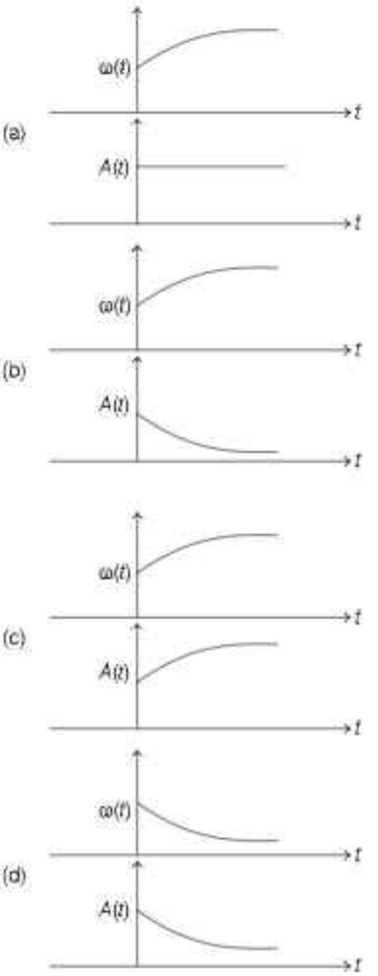
21. Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 min in the direction of her motion, and every 10 min in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

(a) 9 min, 40 km/h (b) 25 min, 100 km/h
(c) 10 min, 90 km/h (d) 15 min, 120 km/h

22. A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (Take $g = 10\ m/s^2$)

(a) 100 N (b) $100\sqrt{3}$ N
(c) 200 N (d) $200\sqrt{3}$ N

23. In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t. Which one of the following options schematically depicts these changes correctly?



24. A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A. It is filled with a gas of density ρ and takes a spherical shape of radius R. When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T. If the speed $v(r)$ of gas coming out of the balloon depends on r as r^α and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

(a) $\alpha = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +\ \delta = \frac{3}{2}$

(b) $\alpha = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$

(c) $\alpha = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

(d) $\alpha = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

25. Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (VSD) are equal to its 9 Main Scale Divisions (MSD). The least division in the MS is 0.1 cm and the zero of VS is at $x = 01$ cm when the jaws of Vernier callipers are closed.

If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

(a) 5.18 cm (b) 5.08 cm
(c) 4.98 cm (d) 5.00 cm

26. A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is

(a) zero at all places
(b) constant between the plates and zero outside the plates.
(c) non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates.
(d) zero between the plates and non-zero outside.

27. An unpolarised light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then

(a) reflected light is completely polarised and the angle of reflection is close to 60°
(b) reflected light is partially polarised and the angle of reflection is close to 30°.
(c) both reflected and transmitted light are perfectly polarised with angles of reflection and refraction close to 60° and 30° respectively.
(d) transmitted light is completely polarised with angle of refraction close to 30°.

28. Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F. A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation, so that for calculating force between them they can be considered as point charges) is best given as

(a) $\frac{3F}{5}$ (b) $\frac{2F}{3}$
(c) $\frac{F}{2}$ (d) $\frac{3F}{8}$

29. A container has two chambers of volumes $V_1 = 2$ L and $V_2 = 3$ L separated by a partition made of a thermal insulator. The chambers contains $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of

(a) 1.3 atm (b) 1.6 atm
(c) 1.4 atm (d) 1.8 atm

30. A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius r of the nth orbit and the particle's speed v in the orbit depend on n as

(a) $r \propto n^{1/3}; v \propto n^{1/3}$ (b) $r \propto n^{1/3}; v \propto n^{2/3}$
(c) $r \propto n^{2/3}; v \propto n^{1/3}$ (d) $r \propto n^{4/3}; v \propto n^{-1/3}$

31. The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury?

(a) 86 earth days (b) 225 earth days
(c) 172 earth days (d) 124 earth days

32. A body weight 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is:

(a) 16 N (b) 27 N
(c) 32 N (d) 36 N

33. A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is

(a) $\frac{R}{64}$ (b) $\frac{R}{32}$
(c) $\frac{R}{16}$ (d) $\frac{R}{8}$

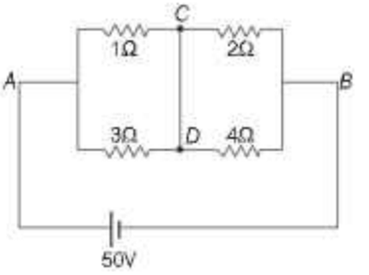
34. de-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

(a) 0.065 nm (b) 0.65 nm
(c) 1.65 nm (d) 2.65 nm

35. An electric dipole with dipole moment 5×10^{-6} Cm is aligned with the direction of a uniform electric field of magnitude 4×10^{5} N/C. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is

(a) 0.8 J (b) 1.0 J
(c) 1.2 J (d) 1.5 J

36. A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is



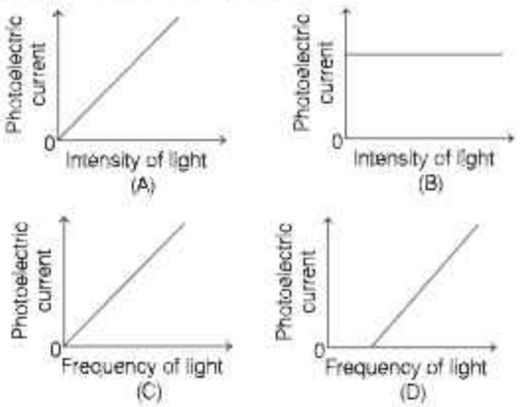
(a) 1.5 A (b) 2.0 A
(c) 2.5 A (d) 3.0 A

37. A photon and an electron (mass m) have the same energy E. The ratio $(\lambda_{photon} / \lambda_{electron})$ of their de-Broglie wavelengths is (c is the speed of light)

(a) $\sqrt{E/2m}$ (b) $c\sqrt{2mE}$

(c) $c\sqrt{\frac{2m}{E}}$ (d) $\frac{1}{c}\sqrt{E/2m}$

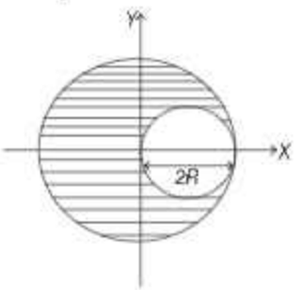
38. Which of the following options represent the variation of photoelectric current with property of light shown on the X-axis?



(a) A only (b) A and C

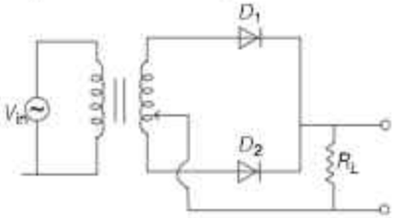
(c) A and D (d) B and D

39. A sphere of radius R is cut from a larger solid sphere or radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is



(a) $\frac{7}{8}$ (b) $\frac{7}{40}$ (c) $\frac{7}{57}$ (d) $\frac{7}{64}$

40. A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{in} = 220 \sin(100\pi)$ V, then at $t = 15$m sec



(a) D_1 is forward biased, D_2 is reverse biased

(b) D_1 is reverse biased, D_2 is forward biased

(c) D_1 and D_2 both are forward biased

(d) D_1 and D_2 both are reverse biased

41. Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio r_A / r_B is equal to

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$

(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{2}$

42. A physical quantity P is related to four observations a, b, c and d as follows:

$$P = a^3 b^2 / c\sqrt{d}$$

The percentage errors of measurement in a, b, c and d are 1%, 3%, 2%, and 4% respectively. The percentage error in the quantity P is

(a) 10% (b) 2%

(c) 13% (d) 15%

43. The intensity of transmitted light when a polaroid sheet, placed between two crossed polarisation at 22.5° from the polarisation axis of one of the polaroid, is (I_0 is the intensity or polarised light after passing through the first polaroid)

(a) $\frac{I_0}{2}$ (b) $\frac{I_0}{4}$

(c) $\frac{I_0}{8}$ (d) $\frac{I_0}{16}$

44. Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2 respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q / A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is

(a) $\frac{k_2}{k_1}$ (b) $\frac{k_1}{k_2}$

(c) $\sqrt{\frac{k_2}{k_1}}$ (d) $\sqrt{\frac{k_1}{k_2}}$

45. A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to

(a) $\frac{f}{2}$ (b) f

(c) $\frac{3f}{2}$ (d) $2f$

CHEMISTRY

46. The ratio of the wavelengths of the light absorbed by a hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow n = 6$ transitions, respectively, is

(a) $\frac{1}{36}$ (b) $\frac{1}{16}$

(c) $\frac{1}{9}$ (d) $\frac{1}{4}$

47. Which of the following statements are true?

A. Unlike Ga that has a very high melting point, Cs has a very low melting point.

B. On Pauling scale, the electronegativity values of N and Cl are not the same.

C. Ar, K^+, Cl^-, Ca^{2+} and S^{2-} are all isoelectronic species.

D. The correct order of the first ionisation enthalpies of Na, Mg, Al, and Si is Si > Al > Mg > Na.

E. The atomic radius of Cs is greater than that of Li and Rb.

Choose the correct answer from the options given below:

(a) A, B and E only (b) C and E only

(c) C and D only (d) A, C and E only

48. Match List I with List II.

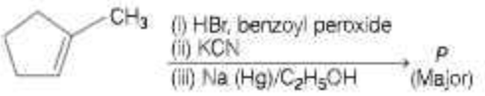
	List I (Ion)		List II (Group number in cation analysis)
A.	Cu^{2+}	I.	Group-I
B.	Mg^{2+}	II.	Group-III
C.	Pb^{2+}	III.	Group-IV
D.	Al^{3+}	IV.	Group-VI

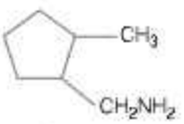
Choose the correct answer from the options given below:

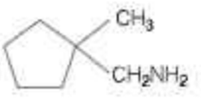
(a) A-III, B-IV, C-II, D-I (b) A-III, B-IV, C-I, D-II

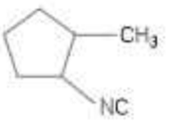
(c) A-III, B-II, C-IV, D-I (d) A-III, B-II, C-I, D-IV

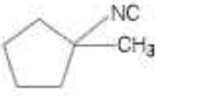
49. Predict the major product 'P' in the following sequence of reactions.



(a) 

(b) 

(c) 

(d) 

50. Energy and radius of first Bohr orbit of He^+ and Li^{2+} are [Given, $R_H = 2.18 \times 10^{-18}$ J, $a_0 = 52.9$ pm].

(a) $E_n(Li^{2+}) = -19.62 \times 10^{-18}$ J; $r_n(Li^{2+}) = 17.6$ pm
$E_n(He^+) = -8.72 \times 10^{-18}$ J; $r_n(He^+) = 26.4$ pm

(b) $E_n(Li^{2+}) = -8.72 \times 10^{-18}$ J; $r_n(Li^{2+}) = 26.4$ pm
$E_n(He^+) = -19.62 \times 10^{-18}$ J; $r_n(He^+) = 17.6$ pm

(c) $E_n(Li^{2+}) = -19.62 \times 10^{-16}$ J; $r_n(Li^{2+}) = 17.6$ pm
$E_n(He^+) = -8.72 \times 10^{-16}$ J; $r_n(He^+) = 26.4$ pm

(d) $E_n(Li^{2+}) = -8.72 \times 10^{-16}$ J; $r_n(Li^{2+}) = 17.6$ pm
$E_n(He^+) = -19.62 \times 10^{-16}$ J; $r_n(He^+) = 17.6$ pm

51. Which of the following are paramagnetic?

A. $[NiCl_4]^{2-}$ B. $Ni(CO)_4$

C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$

E. $Ni(PPh_3)_4$

Choose the correct answer from the options given below:

(a) A and C only (b) B and E only

(c) A and D only (d) A, D and E only

52. Given below are two statements.

Statement I Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both Statement I and Statement II are correct.

(b) Both Statement I and Statement II are incorrect.

(c) Statement I is correct but Statement II is incorrect.

(d) Statement I is incorrect but Statement II is correct.

53. Which among the following electronic configurations belong to main group elements?

A. $[Ne]3s^1$ B. $[Ar]3d^5\,4s^2$

C. $[Kr]4d^{10}5s^25p^5$ D. $[Ar]3d^{10}\,4s^1$

E. $[Rn]5f^0\,6d^2\,7s^2$

Choose the correct answer from the options given below:

(a) B and E only (b) A and C only

(c) D and E only (d) A, C and D only

54. Dalton's atomic theory could not explain which of the following?

(a) Law of conservation of mass

(b) Law of constant proportion

(c) Law of multiple proportion

(d) Law of gaseous volume

55. Consider the following compounds.

$\underline{K}O_2$, $H_2\underline{O}_2$ and $H_2\underline{S}O_4$

The oxidation states of the underlined elements in them are, respectively,

(a) +1, −1, and +6 (b) +2, −2, and +6
(c) +1, −2, and +4 (d) +4, −4, and +6

56. If the half-life ($t_{1/2}$) for a first order reaction is 1 minute, then the time required for 99.9% completion of the reaction is closest to
(a) 2 minutes (b) 4 minutes
(c) 5 minutes (d) 10 minutes

57. The correct order of the wavelength of light absorbed by the following complexes is,

A. $[Co(NH_3)_6]^{3+}$ B. $[Co(CN)_6]^{3-}$

C. $[Cu(H_2O)_4]^{2+}$ D. $[Ti(H_2O)_6]^{3+}$

Choose the correct answer from the options given below:
(a) B < D < A < C (b) B < A < D < C
(c) C < D < A < B (d) C < A < D < B

58. Which one of the following compounds can exist as *cis-trans* isomers?
(a) Pent-1-ene (b) 2-methylhex-2-ene
(c) 1,1-dimethylcyclopropane (d) 1,2-dimethylcyclohexane

59. Phosphoric acid ionises in three steps with their ionisation constant values K_{a_1}, K_{a_2} and K_{a_3}, respectively, while K is the overall ionisation constant. Which of the following statements are true?

A. $\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$

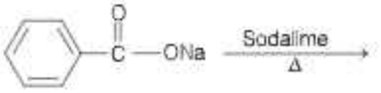
B. H_3PO_4 is stronger acid than $H_2PO_4^-$ and HPO_4^{2-}.

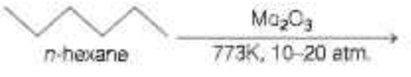
C. $K_{a_1} > K_{a_2} > K_{a_3}$

D. $K_{a_1} = \frac{K_{a_3} + K_{a_2}}{2}$

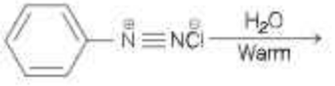
Choose the correct answer from the options given below:
(a) A and B only (b) A and C only
(c) B, C and D only (d) A, B and C only

60. Which one of the following reactions does NOT give benzene as the product?

(a) 

(b) 

(c) 

(d) 

61. If the molar conductivity (Λ_m) of a 0.050 mol L^{-1} solution of a monobasic weak acid is 90 S cm^2 mol^{-1}, its extent (degree) of dissociation will be [Assume, $\Lambda^\circ_+ = 349.6$ S cm^2 mol^{-1} and $\Lambda^\circ_- = 50.4$ S cm^2 mol^{-1}].
(a) 0.115 (b) 0.125 (c) 0.225 (d) 0.215

62. Given below are two statements.

Statement I A hypothetical diatomic molecule with bond order zero is quite stable.

Statement II As bond order increases, the bond length increases.

In the light of the above statements, choose the most appropriate answer from the options given below:
(a) Both statement I and Statement II are true.
(b) Both statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is true but Statement II is true.

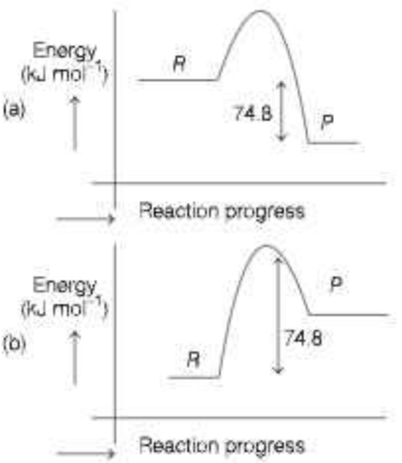
63. Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?
(a) $[Co(NH_3)_3Cl_3]$ (b) $[Co(NH_3)_4Cl_2]$
(c) $[Co(NH_3)_6]Cl_3$ (d) $[Co(NH_3)_5Cl]Cl$

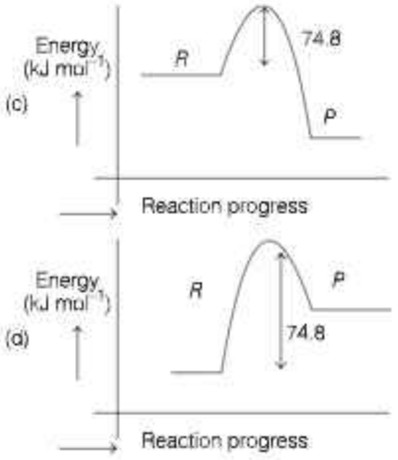
64. Match the List I with List II.

	List I		List II
A.	XeO_3	I.	sp^3d; linear
B.	XeF_2	II.	sp^3; pyramidal
C.	$XeOF_4$	III.	sp^3d^3; distorted octahedral
D.	XeF_6	IV.	sp^3d^2; square pyramidal

Choose the correct answer from the options given below.
(a) A-II, B-I, C-IV, D-III (b) A-II, B-I, C-III, D-IV
(c) A-IV, B-II, C-III, D-I (d) A-IV, B-II, C-I, D-III

65. $C(s) + 2H_2(g) \rightarrow CH_4(g)$; $\Delta H = -74.8$ kJ mol^{-1}.
Which of the following diagrams gives an accurate representation of the above reaction?
[R –> reactants; $P \rightarrow$ products]





66. Match the List I with List II.

	List I (Example)		List II (Type of solution)
A.	Humidity	I.	Solid in solid
B.	Alloys	II.	Liquid in gas
C.	Amalgams	III.	Solid in gas
D.	Smoke	IV.	Liquid in solid

Choose the correct answer from the options given below.

(a) A-II, B-IV, C-I, D-III
(b) A-II, B-I, C-IV, D-III
(c) A-III, B-I, C-IV, D-II
(d) A-III, B-II, C-I, D-IV

67. The correct order of decreasing basic strength of the given amines is

(a) N-methylaniline > benzenamine > ethanamine > N-ethylethanamine.
(b) N-ethylethanamine > ethanamine > benzenamine > N-methylaniline.
(c) N-ethylethanamine > ethanamine > N-methylaniline > benzenamine.
(d) benzenamine > ethanamine > N-methylaniline > N-ethylethanamine.

68. Among the following, choose the ones with equal number of atoms.

A. 212 g of $Na_2CO_3(s)$ [molar mass = 106 g]
B. 248 g of Na_2O (s) [molar mass = 62 g]
C. 240 g of NaOH (s) [molar mass = 40 g]
D. 12 g of $H_2(g)$ [molar mass = 2 g]
E. 220 g of $CO_2(g)$ [molar mass = 44 g]

Choose the correct answer from the options given below:

(a) A, B, and C only
(b) A, B, and D only
(c) B, C, and D only
(d) B, D, and E only

69. Match the List I with List II.

	List I (Name of vitamin)		List II (Deficiency disease)
A.	Vitamin B_{12}	I.	Cheilosis
B.	Vitamin D	II.	Convulsions
C.	Vitamin B_2	III.	Rickets
D.	Vitamin B_6	IV.	Pernicious anaemia

Choose the correct answer from the options given below.

(a) A-I, B-III, C-II, D-IV
(b) A-IV, B-III, C-I, D-II
(c) A-II, B-III, C-I, D-IV
(d) A-IV, B-III, C-II, D-I

70. The correct order of decreasing acidity of the following aliphatic acids is

(a) $(CH_3)_3CCOOH > (CH_3)_2CHCOOH > CH_3COOH > HCOOH$
(b) $CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH > HCOOH$
(c) $HCOOH > CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH$
(d) $HCOOH > (CH_3)_3CCOOH > (CH_3)_2CHCOOH > CH_3COOH$

71. Given below are two statements.

Statement I Ferromagnetism is considered as an extreme form of paramagnetism.

Statement II The number of unpaired electrons in a Cr^{2+} ion ($Z = 24$) is the same as that of a Nd^{3+} ion ($Z = 60$).

In the light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

72. Match the List I with List II.

	List I (Mixture)		List II (Method of separation)
A.	$CHCl_3 + C_6H_5NH_2$	I.	Distillation under reduced pressure
B.	Crude oil in petroleum industry	II.	Steam distillation
C.	Glycerol from spent-lye	III.	Fractional distillation
D.	Aniline-water	IV.	Simple distillation

Choose the correct answer from the options given below.

(a) A-IV, B-III, C-I, D-II
(b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-I, D-II
(d) A-III, B-IV, C-II, D-I

73. For the reaction, $A(g) \rightleftharpoons 2B(g)$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K.

[Given, $R = 0.0831$ L atm $mol^{-1}K^{-1}$]

K_p for the reaction at 1000 K is

(a) 83.1 (b) 2.077×10^5 (c) 0.033 (d) 0.021

74. Given below are two statements.

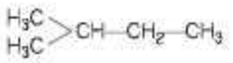
Statement I Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273-278 K. It decomposes easily in the dry state.

Statement II Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the most appropriate answer from the options given below:

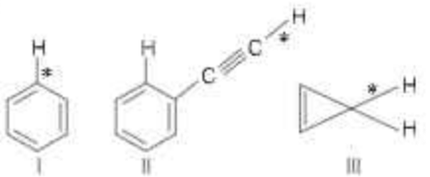
(a) Both Statement I and Statement II are correct.
(b) Both Statement I and Statement II are incorrect.
(c) Statement I is correct but Statement II is incorrect.
(d) Statement I is incorrect but Statement II is correct.

75. How many products (including stereoisomers) are expected from monochlorination of the following compound?



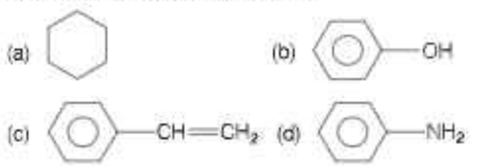
(a) 2 (b) 3 (c) 5 (d) 6

76. Among the given compounds I-III, the correct order of bond dissociation energy of C — H bond marked with * is

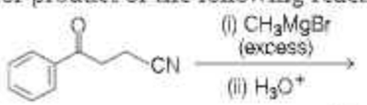


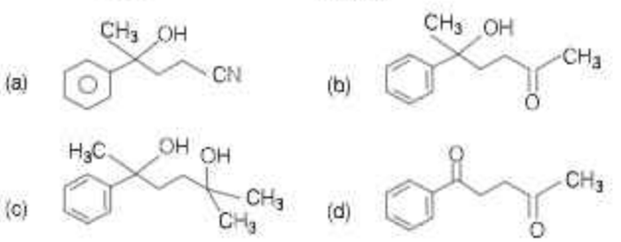
(a) II > I > III (b) I > II > III
(c) III > II > I (d) II > III > I

77. Which one of the following compounds does not decolourise bromine water?



78. The major product of the following reaction is





79. Which of the following aqueous solution will exhibit highest boiling point?

(a) 0.01 M Urea
(b) 0.01 M KNO_3
(c) 0.01 M Na_2SO_4
(d) 0.015 M $C_6H_{12}O_6$

80. Match List I with List II.

	List I		List II
A.	Haber process	I.	Fe catalyst
B.	Wacker oxidation	II.	$PdCl_2$
C.	Wilkinson catalyst	III.	$[(PPh_3)_3RhCl]$
D.	Ziegler catalyst	IV.	$TiCl_4$ with $Al(CH_3)_3$

Choose the correct answer from the options given below:

(a) A-I, B-II, C-IV, D-III
(b) A-II, B-III, C-I, D-IV
(c) A-I, B-II, C-III, D-IV
(d) A-I, B-IV, C-III, D-II

81. 5 moles of liquid X and 10 moles of liquid Y make a solution having a vapour pressure of 70 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Which of the following is true regarding the described solution?

(a) The solution shows positive deviation.
(b) The solution shows negative deviation.
(c) The solution is ideal.
(d) The solution has volume greater than the sum of individual volumes.

82. Sugar 'X'

A. is found in honey.

B. is a keto sugar.

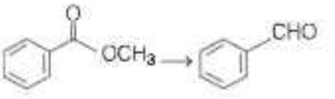
C. exists in α and β-anomeric forms.

D. is laevorotatory.

'X' is

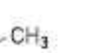
(a) D-glucose (b) D-fructose
(c) Maltose (d) Sucrose

83. Identify the suitable reagent for the following conversion.



(a) (i) $LiAlH_4$, (ii) H^+ / H_2O
(b) (i) $AlH(iBu)_2$, (ii) H_2O
(c) (i) $NaBH_4$, (ii) H^+ / H_2O
(d) (i) H_2 / Pd - $BaSO_4$

84. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A) undergoes S_N2 reaction faster than  .

Reason (R) Iodine is a better leaving group because of its large size.

In the light of the above statements, choose the correct answer from the options given below:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

85. The standard heat of formation, in kcal/mol of Ba^{2+} is

[Given, standard heat of formation of SO_4^{2-} ion $(aq) = -216$ kcal/mol, standard heat of crystallisation of $BaSO_4(s) = -4.5$ kcal/mol, standard heat of formation of $BaSO_4(s) = -349$ kcal/mol].

(a) −128.5 (b) −133.0
(c) +133.0 (d) +220.5

86. Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C_4H_8O is

(a) 6 (b) 8
(c) 10 (d) 11

87. Identify the correct orders against the property mentioned.

A. $H_2O > NH_3 > CHCl_3$ – dipole moment
B. $XeF_4 > XeO_3 > XeF_2$ – number of lone pairs on central atom.
C. O—H > C—H > N—O – bond length
D. $N_2 > O_2 > H_2$ – bond enthalpy

Choose the correct answer from the options given below:

(a) A, D only (b) B, D only
(c) A, C only (d) B, C only

88. Higher yield of NO in

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ can be obtained at

[ΔH of the reaction $= +180.7$ kJ mol^{-1}].

A. Higher temperature
B. Lower temperature
C. Higher concentration of N_2
D. Higher concentration of O_2

Choose the correct answer from the options given below:

(a) A, D only (b) B, C only
(c) B, C, D only (d) A, C, D only

89. If the rate constant of a reaction is 0.03 s^{-1}, how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1}?

(Given, log 2 = 0.301)

(a) 69.3 s (b) 23.1 s (c) 210 s (d) 21.0 s

90. Which one of the following reactions does not belong to "Lassaigne's test"?

(a) $Na + C + N \xrightarrow{\Delta} NaCN$
(b) $2Na + S \xrightarrow{\Delta} Na_2S$
(c) $Na + X \xrightarrow{\Delta} NaX$
(d) $2CuO + C \xrightarrow{\Delta} 2Cu + CO_2$

BIOLOGY

91. The complex-II of mitochondrial electron transport chain is also known as

(a) Cytochrome bc_1
(b) Succinate dehydrogenase
(c) Cytochrome *c* oxidase
(d) NADH dehydrogenase

92. Polymerase chain reaction (PCR) amplifies DNA following the equation.

(a) N^2 (b) 2^n
(c) $2n + 1$ (d) $2N^2$

93. What are the potential drawbacks in adoption of the IVF method?

A. High fatality risk to mother.
B. Expensive instruments and reagents.
C. Husband/wife necessary for being donors.
D. Less adoption of orphans.
E. Not available in india.
F. Possibility that the early embryo does not survive.

Choose the correct answer from the options given below:

(a) B, D, F only (b) A, C, D, F only
(c) A, B, C, D only (d) A, B, C, E, F only

94. What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog?

(a) Aorta
(b) Pulmonary artery
(c) Pulmonary vein
(d) Vena cava

95. Which one of the following statements refers to Reductionist Biology?

(a) Physico-chemical approach to study and understand living organisms.
(b) Physiological approach to study and understand living organisms.
(c) Chemical approach to study and understand living organisms.
(d) Behavioural approach to study and understand living organisms.

96. Given below are two statements.

Statement I In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both statement I and statement II are correct.
(b) Both statement I and statement II are incorrect.
(c) Statement I is correct, but statement II is incorrect.
(d) Statement I is incorrect but statement II is correct.

97. Epiphytes that are growing on a mango branch is an example of which of the following?

(a) Commensalism (b) Mutualism
(c) Predation (d) Amensalism

98. From the statements given below choose the correct option.

A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
B. Each ribosome has two subunits.
C. The two subunits of 80S ribosome are 60S and 40S, while that of 70S are 50S and 30S.
D. The two subunits of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
E. The two subunits of 80S are 60S and 30S and that of 70S are 50S and 30S

(a) A, B, C are true (b) A, B, D are true
(c) A, B, E are true (d) B, D, E are true

99. Which one of the following is an example of ex situ conservation?

(a) National Park (b) Wildlife Sanctuary
(c) Zoos and botanical gardens (d) Protected areas.

100. Given below are two statements:

Statement I The primary source of energy in an ecosystem is solar energy.

Statement II The rate of production of organic matter during photosynthesis in an ecosystem is called net Primary Productivity (NPP).

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct but statement II is incorrect
(d) Statement I is incorrect but statement II is correct

101. Match List-I with List-II.

	List-I		List-II
A.	Emphysema	I.	Rapid spasms in muscle due to low Ca^{++} in body fluid
B.	Angina pectoris	II.	Damaged alveolar walls and decreased respiratory surface
C.	Glomerulo-nephritis	III.	Acute chest pain when not enough oxygen is reaching to heart muscle
D.	Tetany	IV.	Inflammation of glomeruli of kidney

Choose the correct answer from the options given below

(a) A-III, B-I, C-IV, D-II (b) A-III, B-I, C-II, D-IV
(c) A-II, B-IV, C-III, D-I (d) A-II, B-III, C-IV, D-I

102. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Both wind and water pollinated flowers are not very colourful and do not produce nectar.

Reason (R) The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

103. Which of the following is an example of non-distilled alcoholic beverage produced by yeast?

(a) Whisky (b) Brandy
(c) Beer (d) Rum

104. Given below are two statements.

Statement I In a floral formula ⊕ stands for zygomorphic nature of the flower and *G* stands for inferior ovary.

Statement II In a floral formula ⊕ stands for actinomorphic nature of the flower and *G* stands for superior ovary.

In the light of the above statements, choose the most appropriate answer from the options given below:

(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

105. Streptokinase produced by bacterium *Streptococcus* is used for

(a) curd production
(b) ethanol production
(c) liver disease treatment
(d) removing clots from blood vessels

106. Which chromosome in the human genome has the highest number of genes?

(a) Chromosome-X (b) Chromosome-Y
(c) Chromosome-1 (d) Chromosome-10

107. Which of the following statement is correct about location of the male frog copulatory pad?

(a) First and second digit of forelimb
(b) First digit of hindlimb
(c) Second digit of forelimb
(d) First digit of the forelimb

108. Which one of the following phytohormones promotes nutrient mobilisation which helps in the delay of leaf senescence in plants?

(a) Ethylene
(b) Abscisic acid
(c) Gibberellin
(d) Cytokinin

109. While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall, but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelome of that animal?

(a) Acoelomate (b) Pseudocoelomate
(c) Schizocoelomate (d) Spongocoelomate

110. Match List-I with List-II.

	List-I		List-II
A.	Head	I.	Enzymes
B.	Middle piece	II.	Sperm motility
C.	Acrosome	III.	Energy
D.	Tail	IV.	Genetic material

Choose the correct answer from the options given below:

(a) A-IV, B-III, C-I, D-II
(b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I
(d) A-III, B-II, C-I, D-IV

111. Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

A. Prothallus stage
B. Meiosis in spore mother cells
C. Fertilisation
D. Formation of archegonia and antheridia in gametophyte.
E. Transfer of antherozoids to the archegonia in presence of water.

Choose the correct answer from the options given below:

(a) B, A, D, E, C (b) B, A, E, C, D
(c) D, E, C, A, B (d) E, D, C, B, A

112. Cardiac activities of the heart are regulated by

A. nodal tissue.
B. a special neural centre in the medulla oblongata
C. adrenal medullary hormones.
D. adrenal cortical hormones.

Choose the correct answer from the options given below:

(a) A, B and C Only (b) A, B, C and D
(c) A, C and D Only (d) A, B and D Only

113. Which of following organisms cannot fix nitrogen?

A. *Azotobacter* B. *Oscillatoria*
C. *Anabaena* D. *Volvox*
E. *Nostoc*

Choose the correct answer from the options given below.

(a) A only (b) D only
(c) B only (d) E only

114. Given below are two statements.

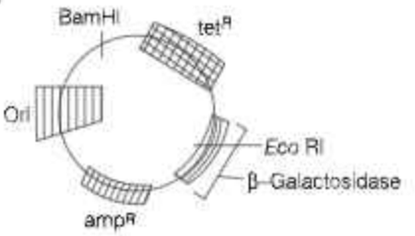
Statement I Transfer RNAs and ribosomal RNA do not interact with *m*RNA.

Statement II RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

115.



In the above represented plasmid an alien piece of DNA is inserted at *Eco*RI site. Which of the following strategies will be chosen to select the recombinant colonies?

(a) Using ampicillin and tetracyclin containing medium plate.
(b) Blue colour colonies will be selected.
(c) White colour colonies will be selected.
(d) Blue color colonies grown on ampicillin plates can be selected.

116. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

(a) Bacterium (b) Yeast
(c) Virus (d) Phage

117. Name the class of enzyme that usually catalyse the following reaction,

$$S—G + S'' \rightarrow S + S''—G$$

Where, G → a group other than hydrogen
S → a substrate
S'' → another substrate

(a) Hydrolase (b) Lyase
(c) Transferase (d) Ligase

118. Find the statement that is not correct with regard to the structure of monocot stem.

(a) Hypodermis is parenchymatous.
(b) Vascular bundles are scattered.
(c) Vascular bundles are conjoint and closed.
(d) Phloem parenchyma is absent.

119. The correct sequence of events in the life cycle of bryophytes is

A. fusion of antherozoid with egg.
B. attachment of gametophyte to substratum.
C. reduction division to produce haploid spores.
D. formation of sporophyte.
E. release of antherozoids into water.

Choose the correct answer from the option given below:
(a) D, E, A, C, B (b) B, E, A, C, D
(c) B, E, A, D, C (d) D, E, A, B, C

120. Which are correct.
A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.
B. Chemotherapeutics drugs are used to kill non-cancerous cells.
C. α-interferon activate the cancer patients immune system and helps in destroying the tumour.
D. Chemotherapeutic drugs are biological response modifiers.
E. In the case of leukaemia blood cells counts are decreased.

Choose the correct answer from the option given below:
(a) B and D only (b) D and E only
(c) C and D only (d) A and C only

121. Match List I with List II.

	List I		List II
A.	Centromere	I.	Mitochondrion
B.	Cilium	II.	Cell division
C.	Cristae	III.	Cell movement
D	Cell membrane	IV.	Phospholipid bilayer

Choose the correct answer from the options given below.
(a) A-I, B-II, C-III, D-IV (b) A-II, B-I, C-IV, D-III
(c) A-IV, B-II, C-III, D-I (d) A-II, B-III, C-I, D-IV

122. Match List I with List II.

	List I		List II
A.	Chlorophyll-*a*	I.	Yellow-green
B.	Chlorophyll-*b*	II.	Yellow
C.	Xanthophylls	III.	Blue-green
D	Carotenoids	IV.	Yellow to Yellow-orange

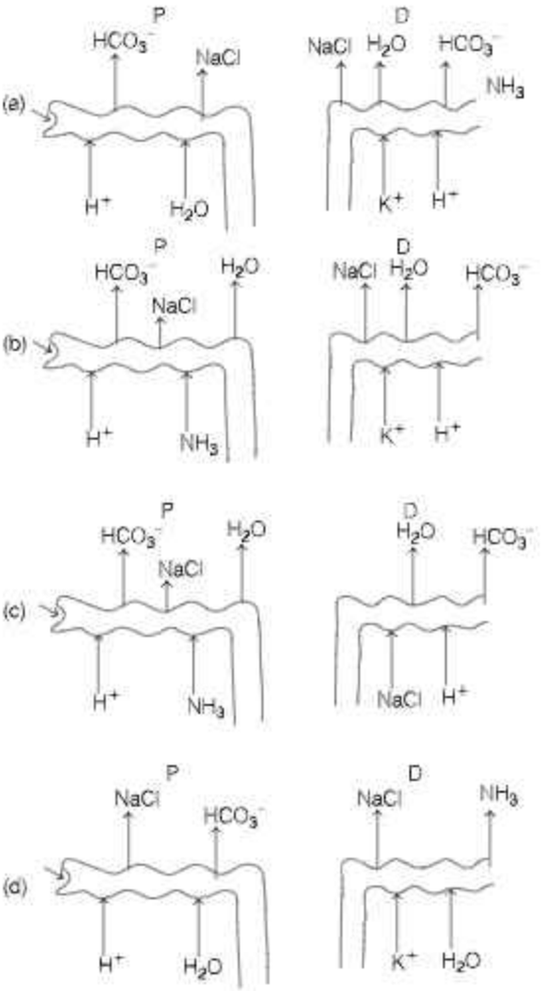
Choose the option with all correct matches.
(a) A-III, B-IV, C-II, D-I (b) A-III, B-I, C-II, D-IV
(c) A-I, B-II, C-IV, D-III (d) A-I, B-IV, C-III, D-II

123. Find the correct statements.
A. In human pregnancy, the major organ systems are formed at the end of 12 weeks.
B. In human pregnancy, the major organ systems are formed at the end of 8 weeks.
C. In human pregnancy, heart is formed after one month of gestation.
D. In human pregnancy, limbs and digits develop by the end of second month.
E. In human pregnancy the appearance of hair usually observed in the fifth month.

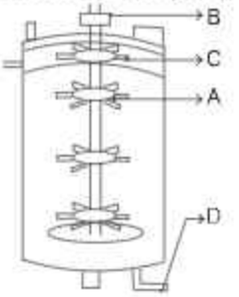
Choose the correct answer from the options given below.
(a) A and E only (b) B and C only
(c) B, C, D and E only (d) A, C, D and E only

124. In the seeds of cereals, the outer covering of endosperm separates the embryo by a protein-rich layer called:
(a) Coleoptile (b) Coleorhiza
(c) Integument (d) Aleurone layer

125. Which of the following diagrams is correct with regard to the proximal (P) and distal (D) tubule of the Nephron.



126. Identify the part of a bio-reactor which is used as a foam braker from the given figure.



(a) A (b) B
(c) D (d) C

127. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) A typical unfertilised, angiosperm, embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R) The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

128. A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is:

(a) mesosome (b) chromatophores
(c) cristae (d) endoplasmic reticulum

129. Which of the following are the post-transcriptional events in an eukaryotic cell?

A. Transport of pre-*m*RNA to cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at 5′ end of hnRNA.
D. Addition of adenine residues at 3′ end of hnRNA.
E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below.

(a) A, B, C only (b) B, C, D only
(c) B, C, E only (d) C, D, E only

130. What is the pattern of inheritance for polygenic trait?

(a) Mendelian inheritance pattern
(b) Non-mendelian inheritance pattern
(c) Autosomal dominant pattern
(d) X-linked recessive inheritance pattern

131. Which one of the following enzymes contains 'Haem' as the prosthetic group?

(a) RuBisCo
(b) Carbonic anhydrase
(c) Succinate dehydrogenase
(d) Catalase

132. Each of the following characteristics represent a kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organisation.

A. Multicellular heterotrophs with cell wall made of chitin.
B. Heterotrophs with tissue/organ/organ system level of body organisation.
C. Prokaryotes with cell wall made of polysaccharides and amino acids.
D. Eukaryotic autotrophs with tissue/organ level of body organisation.
E. Eukaryotes with cellular body organisation.

Choose the correct answer from the options given below.

(a) A, C, E, B, D (b) C, E, A, D, B
(c) A, C, E, D, B (d) C, E, A, B, D

133. Who is known as the father of ecology in India?

(a) S. R. Kashyap (b) Ramdeo Misra
(c) Ram Udar (d) Birbal Sahni

134. Match List-I with List-II.

	List-I		List-II
A.	Alfred Hershey and Martha Chase	I.	*Streptococcus pneumoniae*
B.	Euchromatin	II.	Densely packed and dark-stained
C.	Frederick Griffith	III.	Loosely packed and light-stained
D.	Heterochromatin	IV.	DNA as genetic material confirmation

Choose the correct answer from the options given below.

(a) A-II, B-IV, C-I, D-III (b) A-IV, B-II, C-I, D-III
(c) A-IV, B-III, C-I, D-II (d) A-III, B-II, C-IV, D-I

135. Neoplastic characteristics of cells refers to:

A. A mass of proliferating cell
B. Rapid growth of cells
C. Invasion and damage to the surrounding tissue
D. Those confined to original location

Choose the correct answer from the options given below.

(a) A, B only (b) A, B, C only
(c) A, B, D only (d) B, C, D only

136. Given below are two statements:

Statement I The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II Smaller size DNA fragments are observed near anode, while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

137. Match List I with List II.

	List I		List II
A.	Adenosine	I.	Nitrogen base
B.	Adenylic acid	II.	Nucleotide
C.	Adenine	III.	Nucleoside
D.	Alanine	IV.	Amino acid

Choose the option with all correct matches.
(a) A-III, B-IV, C-II, D-I
(b) A-III, B-II, C-IV, D-I
(c) A-III, B-II, C-I, D-IV
(d) A-II, B-III, C-I, D-IV

138. Consider the following.
A. The reductive division for the human female gametogenesis starts earlier than that of the male gametogenesis.
B. The gap between the first meiotic division and the second meiotic division is much shorter for males compared to females.
C. The first polar body is associated with the formation of the primary oocyte.
D. Luteinizing Hormone (LH) surge leads to disintegration of the endometrium and onset of menstrual bleeding.

Choose the correct answer from the options given below.
(a) A and B are true (b) A and C are true
(c) B and D are true (d) B and C are true

139. All living members of the class Cyclostomata are:
(a) free-living (b) endoparasite
(c) symbiotic (d) ectoparasite

140. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R) Vesicles containing materials made by the endoplasmic reticulum fuse with the *cis* face of the Golgi apparatus and they are modified and released from the *trans* face of the Golgi apparatus.

In the light of the above statements, choose the correct answer from the options given below
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

141. Match List I with List II.

	List I		List II
A.	Scutellum	I.	Persistent nucellus
B.	Non-albuminous seed	II.	Cotyledon of Monocot seed
C.	Epiblast	III.	Groundnut
D.	Perisperm	IV.	Rudimentary cotyledon

Choose the option with all correct matches.
(a) A-II, B-III, C-IV, D-I
(b) A-IV, B-III, C-II, D-I
(c) A-IV, B-III, C-I, D-II
(d) A-II, B-IV, C-III, D-I

142. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) All vertebrates are chordates, but all chordates are not vertebrate.

Reason (R) The members of subphylum vertebrata possess notochord during the embryonic period, the notochord is replaced by a cartilaginous or bony vertebral column in adults.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

143. Identify the statement that is not correct.
(a) Each antibody has two light and two heavy chains.
(b) The heavy and light chains are held together by disulfide bonds.
(c) Antigen binding site is located at C-terminal region of antibody molecules.
(d) Constant region of heavy and light chains are located at C-terminus of antibody molecules.

144. Silencing of specific mRNA is possible *via* RNAi because of
(a) complementary *ds*RNA (b) inhibitory *ss*RNA
(c) complementary *t*RNA
(d) non-complementary *ss*RNA

145. Genes R and Y follow independent assortment. If RRYY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F_2-generation?
(a) Phenotypic ratio - 1 : 2 : 1
(b) Phenotypic ratio - 3 : 1
(c) Phenotypic ratio - 9 : 3 : 3 : 1
(d) Phenotypic ratio - 9 : 7

146. Histones are enriched with -
(a) lysine and arginine
(b) leucine and lysine
(c) phenylalanine and leucine
(d) phenylalanine and arginine

147. The first menstruation is called
(a) menopause (b) menarche
(c) diapause (d) ovulation

148. Match List I with List II.

	List I		List II
A.	Heart	I.	Erythropoietin
B.	Kidney	II.	Aldosterone
C.	Gastro-intestinal tracts	III.	Atrial natriuretic factor
D.	Adrenal cortex	IV.	Secretin

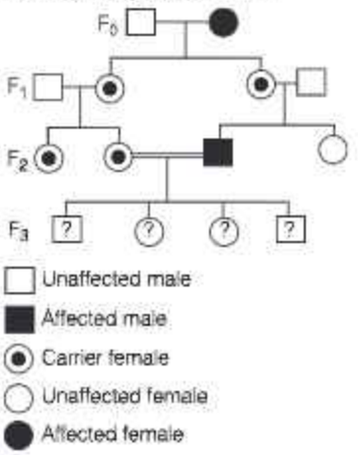
Choose the correct answer from the options given below:
(a) A-II, B-I, C-III, D-IV (b) A-IV, B-III, C-II, D-I
(c) A-I, B-III, C-IV, D-II (d) A-III, B-I, C-IV, D-II

149. The protein portion of an enzyme is called
(a) cofactor (b) coenzyme
(c) apoenzyme (d) prosthetic group

150. Which of the following is the unit of productivity of an ecosystem?
(a) gm^{-2} (b) kcal m^{-2}
(c) kcal m^{-3} (d) (kcal m^{-2})yr^{-1}

151. Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.
(a) Analogy, convergent (b) Homology, divergent
(c) Homology, convergent (d) Analogy, divergent

152. With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F_3-generation.



(a) 1/4 (b) 1/2
(c) 1/8 (d) Zero

153. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

Reason (R) Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

154. How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?
(a) 2 meiosis and 3 mitosis
(b) 1 meiosis and 2 mitosis
(c) 1 meiosis and 3 mitosis
(d) No meiosis and 2 mitosis

155. Which of the following is an example of a zygomorphic flower?
(a) *Petunia* (b) *Datura*
(c) Pea (d) Chilli

156. After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s)/tissue(s) like:
A. thymus B. bone marrow
C. spleen D. lymph nodes
E. Peyer's patches
Choose the correct answer from the options given below.
(a) B, C, D only (b) A, B, C only
(c) E, A, B only (d) C, D, E only

157. Given below are two statements.

Statement I Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

158. What is the main function of the spindle fibers during mitosis?
(a) To separate the chromosomes
(b) To synthesise new DNA
(c) To repair damaged DNA
(d) To regulate cell growth

159. Which one of the following is the characteristic feature of gymnosperms?
(a) Seeds are enclosed in fruits.
(b) Seeds are naked.
(c) Seeds are absent.
(d) Gymnosperms have flowers for reproduction.

160. Consider the following statements regarding function of adrenal medullary hormones.
A. It causes pupillary constriction
B. It is a hyperglycemic hormone
C. It causes piloerection
D. It increases/strength of heart contraction
Choose the correct answer from the options given below:
(a) C and D Only
(b) B, C and D Only
(c) A, C and D Only
(d) D Only

161. Why can't insulin be given orally to diabetic patients?
(a) Human body will elicit strong immune response
(b) It will be digested in Gastro-Intestinal (GI) tract
(c) Because of structural variation
(d) Its bioavailability will be increased

162. Match List I with List II.

	List I		List II
A.	Pteridophyte	I.	*Salvia*
B.	Bryophyte	II.	*Ginkgo*
C.	Angiosperm	III.	*Polytrichum*
D.	Gymnosperm	IV.	*Salvinia*

Choose the option with all correct matches.
(a) A-III, B-IV, C-II, D-I (b) A-IV, B-III, C-I, D-II
(c) A-III, B-IV, C-I, D-II (d) A-IV, B-III, C-II, D-I

163. Who proposed that the genetic code for amino acids should be made up of three nucleotides?
(a) George Gamow (b) Francis Crick
(c) Jacque Monod (d) Franklin Stahl

164. Match List I with List II.

	List I		List II
A.	The Evil Quartet	I.	Cryopreservation
B.	*Ex situ* conservation	II.	Alien species invasion
C.	Lantana camara	III.	Causes of biodiversity losses
D.	*Dodo*	IV.	Extinction

Choose the option with all correct matches.
(a) A-III, B-II, C-I, D-IV (b) A-III, B-I, C-II, D-IV
(c) A-III, B-IV, C-II, D-I (d) A-III, B-II, C-IV, D-I

165. Which of the following hormones released from the pituitary is actually synthesised in the hypothalamus?
(a) Luteinizing Hormone (LH)
(b) Anti-Diuretic Hormone (ADH)
(c) Follicle-Stimulating Hormone (FSH)
(d) Adenocorticotrophic Hormone (ACTH)

166. Role of the water vascular system in Echinoderms is
A. respiration and locomotion
B. excretion and locomotion
C. capture and transport of food
D. digestion and respiration
E. digestion and excretion
Choose the correct answer from the options given below.
(a) A and B Only (b) A and C Only
(c) B and C Only (d) B, D and E Only

167. Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body?
(a) Acquired immunity (b) Innate immunity
(c) Cell-mediated immunity (d) Humoral immunity

168. In bryophytes, the gemmae help in which one of the following?
(a) Sexual reproduction (b) Asexual reproduction
(c) Nutrient absorption (d) Gaseous exchange

169. In frog, the renal portal system is a special venous connection that acts to link :
(a) liver and intestine (b) liver and kidney
(c) kidney and intestine
(d) kidney and lower part of body

170. Given below are two statements.
Statement I In ecosystem, there is unidirectional flow of energy of sun from producers to consumers.
Statement II Ecosystems are exempted from 2nd law of thermodynamics.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

171. Which of the following statements about RuBisCO is true?
(a) It is active only in the dark.
(b) It has higher affinity for oxygen than carbon dioxide.
(c) It is an enzyme involved in the photolysis of water.
(d) It catalyses the carboxylation of RuBP.

172. Which of the following enzyme(s) are not essential for gene cloning?
A. Restriction enzymes B. DNA ligase
C. DNA mutase
D. DNA recombinase
E. DNA polymerase
Choose the correct answer from the options given below.
(a) C and D only (b) A and B only
(c) D and E only (d) B and C only

173. Read the following statements on plant growth and development.
A. Parthenocarpy can be induced by auxins.
B. Plant growth regulators can be involved in promotion as well as inhibition of growth.
C. Dedifferentiation is a pre-requisite for re-differentiation.
D. Abscisic acid is a plant growth promoter.
E. Apical dominance promotes the growth of lateral buds.
Choose the option with all correct statements.
(a) A, B, C only (b) A, C, E only
(c) A, D, E only (d) B, D, E only

174. Which factor is important for termination of transcription?
(a) α (alpha) (b) σ (sigma) (c) ρ (rho) (d) γ (gamma)

175. Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs.
Choose the correct answer from the following.
(a) The statement is true for water, but false for land
(b) The statement is true for both the environment
(c) The statement is false for water, but true for land
(d) The statement is false for both the environment

176. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?
(a) They are monozygotic twins.
(b) They are fraternal twins.
(c) They were conceived through *in vitro fertilisation*.
(d) They have 75% identical genetic content.

177. Which of the following microbes is not involved in the preparation of household products?

A. *Aspergillus niger*

B. *Lactobacillus*

C. *Trichoderma polysporum*

D. *Saccharomyces cerevisiae*

E. *Propionibacterium shermanii*

Choose the correct answer from the options given below.

(a) A and B only (b) A and C only
(c) C and D only (d) C and E only

178. Match List-I with List-II.

List I	List II
A. Progesterone	I. Pars intermedia
B. Relaxin	II. Ovary
C. Melanocyte stimulating hormone	III. Adrenal medulla
D. Catecholamines	IV. Corpus luteum

Choose the correct answer from the options given below:

(a) A-IV, B-II, C-I, D-III (b) A-IV, B-II, C-III, D-I
(c) A-II, B-IV, C-I, D-III (d) A-III, B-II, C-IV, D-I

179. The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method.

Statement I The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

180. Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

(a) $\frac{dN}{dt} = r\left(\frac{K-N}{K}\right)$ (b) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(c) $\frac{dN}{dt} = rN\left(\frac{N-K}{K}\right)$ (d) $\frac{dN}{dt} = N\left(\frac{r-N}{K}\right)$

Answers

Physics

1. (b)	2. (b)	3. (b)	4. (d)	5. (b)	6. (b)	7. (d)	8. (d)	9. (a)	10. (a)
11. (d)	12. (a)	13. (c)	14. (c)	15. (b)	16. (b)	17. (d)	18. (b)	19. (c)	20. (a)
21. (d)	22. (b)	23. (b)	24. (c)	25. (c)	26. (c)	27. (a)	28. (d)	29. (b)	30. (c)
31. (a)	32. (b)	33. (c)	34. (b)	35. (b)	36. (b)	37. (c)	38. (a)	39. (c)	40. (b)
41. (b)	42. (c)	43. (c)	44. (d)	45. (b)					

Chemistry

46. (d)	47. (b)	48. (b)	49. (a)	50. (a)	51. (c)	52. (c)	53. (b)	54. (d)	55. (a)
56. (d)	57. (b)	58. (d)	59. (d)	60. (d)	61. (c)	62. (b)	63. (a)	64. (a)	65. (a)
66. (b)	67. (c)	68. (b)	69. (b)	70. (c)	71. (c)	72. (a)	73. (c)	74. (a)	75. (d)
76. (a)	77. (a)	78. (b)	79. (c)	80. (c)	81. (b)	82. (b)	83. (b)	84. (a)	85. (a)
86. (c)	87. (a)	88. (d)	89. (a)	90. (d)					

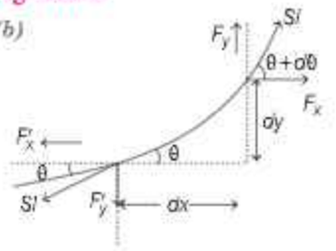
Biology

91. (b)	92. (b)	93. (a)	94. (d)	95. (a)	96. (a)	97. (a)	98. (a)	99. (c)	100. (c)
101. (d)	102. (b)	103. (c)	104. (d)	105. (d)	106. (c)	107. (d)	108. (d)	109. (b)	110. (a)
111. (a)	112. (a)	113. (b)	114. (d)	115. (c)	116. (a)	117. (c)	118. (a)	119. (c)	120. (d)
121. (d)	122. (b)	123. (d)	124. (d)	125. (b)	126. (d)	127. (c)	128. (a)	129. (b)	130. (b)
131. (d)	132. (b)	133. (b)	134. (c)	135. (b)	136. (a)	137. (c)	138. (a)	139. (d)	140. (a)
141. (a)	142. (a)	143. (c)	144. (a)	145. (c)	146. (a)	147. (b)	148. (d)	149. (c)	150. (d)
151. (a)	152. (a)	153. (a)	154. (c)	155. (c)	156. (d)	157. (b)	158. (a)	159. (b)	160. (b)
161. (b)	162. (b)	163. (a)	164. (b)	165. (b)	166. (b)	167. (b)	168. (b)	169. (d)	170. (c)
171. (d)	172. (a)	173. (a)	174. (c)	175. (c)	176. (b)	177. (b)	178. (a)	179. (d)	180. (b)

Answer with Explanations

Physics

1. *(b)*



For the given element, consider length l in z-directions

Net force in upward Direction = Weight

$$[S\sin(\theta + d\theta) - S\sin\theta]l = dmg$$

Here, angle is small

$$\therefore \quad \sin\theta = \theta \Rightarrow Sld\theta = dmg$$

$$\Rightarrow \quad Sld\theta = (\rho ydxl)g \quad (\because dm = \rho ydxl)$$

$$\Rightarrow \quad \frac{d\theta}{ydx} = \frac{\rho g}{S} \qquad \ldots(i)$$

$$\tan\theta = dy/dx$$

Differentiating w.r.t. x

$$\sec^2\theta\frac{d\theta}{dx} = \frac{d^2y}{dx^2} \qquad \ldots(ii)$$

Put $d\theta$ from Eq. (ii) in Eq. (i) and take $\cos\theta \simeq 1$, we get

$$\frac{d^2y}{dx^2} = \frac{\rho gy}{S}$$

2. *(b)* $f_0 = 2$ cm, $f_e = 4$ cm, $L = 40$ cm, $D = 25$ cm

In the condition of normal adjustment

$$m = \frac{L}{f_0}\cdot\frac{D}{f_e} = \frac{40}{2}\times\frac{25}{4} = 125$$

Correct answer is (b).

3. *(b)* For particle to remain undeflected,

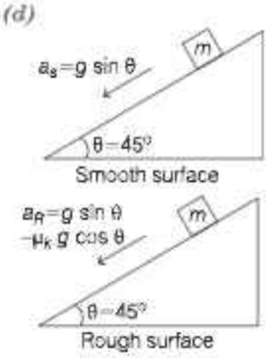
$$\mathbf{F}_e = \mathbf{F}_m \Rightarrow |\mathbf{E}| = vB\sin\theta$$

$$E = \frac{c}{100}\times 9\times10^{-4}\sin 90^\circ$$

$$E = 27\times10^5 \text{ V/m and}$$

$$\mathbf{E} = \mathbf{v}\times\mathbf{B} \; i.e., \; E \perp B$$

4. *(d)*



$$L = \frac{1}{2}a_S t_s^2 = \frac{1}{2}a_R t_R^2$$

$$\left(\frac{t_S}{t_R}\right)^2 = \frac{a_R}{a_S}$$

$$\Rightarrow \left(\frac{1}{2}\right)^2 = \frac{g\sin\theta - \mu_k g\cos\theta}{g\sin\theta}$$

$$\Rightarrow \quad \frac{1}{4} = \frac{\sin 45^\circ - \mu_k\cos 45^\circ}{\sin 45^\circ}$$

$$\frac{1}{4} = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\left(\frac{1}{\sqrt{2}}\right)}\left(\frac{1-\mu_k}{1}\right) \Rightarrow \frac{1}{4} = 1 - \mu_k$$

$$\Rightarrow \quad \mu_k = \frac{3}{4} = 0.75$$

5. *(b)* Work done = Change in KE

$$\mathbf{F}\cdot\mathbf{s} = \Delta KE$$

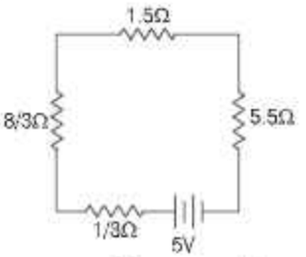
$$\frac{(\Delta KE)_A}{(\Delta KE)_B} = \frac{-F_A(s_A)}{-F_B(s_B)}$$

$$\Rightarrow \frac{-100}{-225} = \frac{-F_A(1000)}{-F_B(1500)}$$

$$\frac{F_A}{F_B} = \frac{2}{3}$$

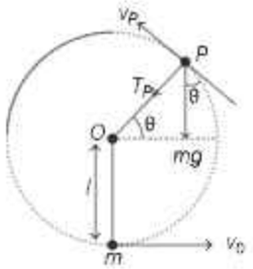
6. *(b)* In the given diagram, remove the 6Ω resistor because point A and D are at equal potential, $\frac{5}{2.5} = \frac{3}{1.5}$.

Now, the simplied diagram is shown below.



$$I = \frac{5}{\frac{1}{3} + \frac{8}{3} + 1.5 + 5.5} = \frac{5}{10} = 0.5 \text{ A}$$

7. *(d)*



Conservation of mechanical energy

$$\frac{1}{2}mv_0^2 = mgl(1+\sin\theta) + \frac{1}{2}mv_p^2 \qquad \ldots(i)$$

At point P

$$T_p + mg\sin\theta = \frac{mv_p^2}{l} \quad (\text{as } T_p = 0)$$

$$mg\sin\theta = \frac{mv_p^2}{l}$$

$$\Rightarrow \quad mv_p^2 = mgl\sin\theta \qquad \ldots(ii)$$

From Eqs. (i) and (ii)

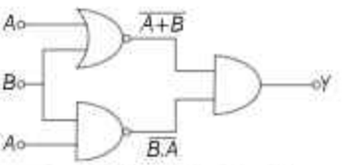
$$\frac{1}{2}mv_0^2 = mgl(1+\sin\theta) + \frac{1}{2}mgl\sin\theta$$

$$v_0^2 = 2gl(1+\sin\theta) + gl\sin\theta$$

$$v_0 = \sqrt{2gl + 3gl\sin\theta}$$

and $\quad v_p = \sqrt{gl\sin\theta} \Rightarrow \frac{v_p}{v_0} = \sqrt{\frac{\sin\theta}{2+3\sin\theta}}$

8. *(d)*



$$Y = (\overline{A+B})\cdot(\overline{A\cdot B}) = \overline{A}\cdot\overline{B}(\overline{A}+\overline{B})$$

$$= \overline{A}\cdot\overline{B}\cdot\overline{A} + \overline{A}\cdot\overline{B}\cdot\overline{B} = \overline{A}\cdot\overline{B} + \overline{A}\cdot\overline{B}$$

$$= \overline{A}\cdot\overline{B} = \overline{(A+B)}$$

Behaves as NOR gate.

9. *(a)* $v = \frac{\omega}{k} = \frac{1.5\times10^8}{5} = 3\times10^8$ m/s

$$B_0 = \frac{E_0}{c} = \frac{60}{3\times10^8} = 20\times10^{-8} = 2\times10^{-7}\text{T}$$

Also, magnetic field must be normal to both E and x i.e., magnetic field must be in y direction.

10. *(a)* $v_i = -\sqrt{2gh_i} = -\sqrt{2\times9.8\times40}$

$$= -28 \text{ m/s}$$

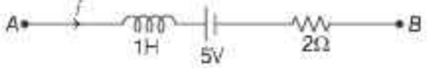
$$v_f = \sqrt{2gh_f} = \sqrt{2\times9.8\times10} = 14 \text{ m/s}$$

$$\mathbf{I} = \Delta\mathbf{p} = m\mathbf{v}_f - m\mathbf{v}_i$$

$$= m[\mathbf{v}_f - \mathbf{v}_i] = 0.5[14 - (-28)]$$

$$= 0.5\times42 = 21 \text{ Ns}$$

11. *(d)*



Applying KVL

$$V_A - \frac{Ldi}{dt} - 5 - IR = V_B$$

$$V_A - V_B = \frac{Ldi}{dt} + 5 + IR$$

$$= 1\times1 + 5 + 2\times2 = 10\text{V}$$

12. *(a)* Magnetic moment $M = IA$

$$M = I(\pi r^2)$$

$$\frac{M_1}{M_2} = \frac{I\pi r_1^2}{I\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{1}{2}\right)^2$$

$$\Rightarrow \quad \frac{M_1}{M_2} = \frac{1}{4}$$

13. *(c)* When a combination of lens is used then total power

$$P_{net} = P + P + P + P = 4P$$

Total magnification

$$M_{net} = m \times m \times m \times m = m^4$$

14. *(c)* $n_f = \dfrac{p_f V}{RT}$

$[p_f = p_{tot} = p_{oxygen} + p_{atm} = 11 + 1 = 12 \text{ atm}]$

$$= \frac{12 \times 1.01 \times 10^5 \times 30 \times 10^{-3}}{\left(\frac{100}{12}\right) \times 300}$$

$$= 14.544$$

$\therefore$ Moles removed $= 18.2 - 14.544 = 3.656$

$\therefore$ Mass removed

$$= 3.656 \times \frac{32}{1000}\text{kg} = 0.116\text{kg}$$

15. *(b)* $t = x^3 + x$

$$\frac{dt}{dx} = \frac{d}{dx}[x^3 + x]$$

$$\frac{1}{v} = 2x + 1$$

$$\Rightarrow \quad v = (2x+1)^{-1} \qquad \ldots(i)$$

$$a = \frac{v dv}{dx} = (2x+1)^{-1}[-1(2x+1)^{-2} \times 2]$$

$$a = -1(2x+1)^{-3} \times 2$$

$$a = -2(2x+1)^{-3} = -\frac{2}{(2x+1)^3}$$

16. *(b)* $X_C = 25\ \Omega \quad X_L = 45\Omega$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = \sqrt{20^2 + (45-25)^2} \Rightarrow Z = 20\sqrt{2}$$

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{220}{20\sqrt{2}} = \frac{11}{\sqrt{2}} = 7.8\text{ A}$$

$$\tan\phi = \frac{X_L - X_C}{R} = \frac{45-25}{20} = \frac{20}{20} = 1$$

$$\phi = 45^\circ$$

17. *(d)* M = same

$$R_2 = 2R_1$$

$$I_{sphere} \propto r^2$$

$$I_2 = 4I_1$$

Using angular momentum conservation,

$$I_1\omega_1 = I_2\omega_2 \Rightarrow \omega_2 = \omega_1 / 4$$

Now, $T = \dfrac{2\pi}{\omega}$

$$\Rightarrow \quad T_2 = 4T_1 = 4 \times 27 \quad (\because T_1 = 27 \text{ days})$$

$$T_2 = 108 \text{ days}$$

18. *(b)* Magnetic moment,

$$M = IA = I(\pi r^2)$$

$$M = \left(\frac{ev}{2\pi r}\right)(\pi r^2) \qquad \ldots(i)$$

Given, $B(\pi r^2) = n\left(\dfrac{h}{e}\right)$

$$\Rightarrow \quad r^2 = \frac{h}{B\pi e} \qquad \ldots(ii) \quad (\because n = 1)$$

and when charge is moving in external magnetic field then, $r = \dfrac{mv}{qB}$

$$\frac{v}{r} = \frac{eB}{m} \qquad \ldots(iii) \quad (\because q = e)$$

Put value from Eq. (ii) and (iii) in Eq. (i)

$$M = \left(\frac{ev}{2\pi r}\right)(\pi r^2)$$

$$M = \frac{e}{2\pi}\left(\frac{eB}{m}\right)\pi\left(\frac{h}{B\pi e}\right) \Rightarrow M = \frac{eh}{2\pi m}$$

19. *(c)* In steady state

$$\left(\frac{dQ}{dt}\right)_1 = \left(\frac{dQ}{dt}\right)_2 = \left(\frac{dQ}{dt}\right)_3$$

$$\frac{2KA(3T - T_1)}{L} = \frac{KA(T_1 - T_2)}{L} = \frac{2KA(T_2 - T)}{L}$$

$$6T - 2T_1 = T_1 - T_2 = 2T_2 - 2T$$

Now, $6T - 2T_1 = T_1 - T_2$

$$6T + T_2 = 3T_1 \qquad \ldots(i)$$

$$T_1 - T_2 = 2T_2 - 2T$$

$$2T - 3T_2 = -T_1 \qquad \ldots(ii)$$

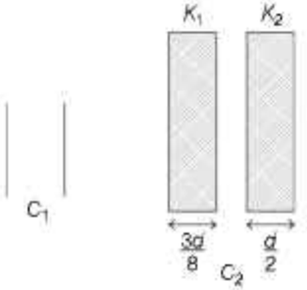
Multiplying Eq. (ii) by 3

$$6T - 9T_2 = -3T_1 \qquad \ldots(iii)$$

Eq. (i) – Eq. (iii)

$$10T_2 = 6T_1 \Rightarrow \frac{T_1}{T_2} = \frac{5}{3}$$

20. *(a)* $C_1 = \dfrac{\varepsilon_0 A}{d}$



$$C_2 = \frac{\varepsilon_0 A}{\frac{3d}{8K_1} + \frac{d}{2K_2} + \left(d - \frac{3d}{8} - \frac{d}{2}\right)}$$

$$C_2 = \frac{\frac{\varepsilon_0 A}{d}}{\frac{3}{8K_1} + \frac{1}{2K_2} + \frac{1}{8}}$$

$$C_2 = 2C_1 \qquad \text{(given)}$$

$$\frac{\frac{\varepsilon_0 A}{d}}{\frac{3}{8K_1} + \frac{5}{8K_1} + \frac{1}{8}} = \frac{2\varepsilon_0 A}{d}$$

$$\left(\because K_2 = 1.25\,K_1 = \frac{5}{4}K_1\right)$$

$$\frac{1}{\frac{1}{K_1} + \frac{1}{8}} = 2 \Rightarrow \frac{1}{K_1} + \frac{1}{8} = \frac{1}{2}$$

$$\Rightarrow \quad K_1 = \frac{8}{3} = 2.66$$

21. *(d)* City - X, City - Y

$t_1 = 30\text{min} = \dfrac{1}{2}\text{h}, \qquad t_2 = 10\text{min} = \dfrac{1}{6}\text{h}$

v_B = Speed of bus,

v_S = Speed of scooty (girl)

$$d = (v_B - v_S)t_1 = (v_B + v_S)t_2$$

$$\Rightarrow (v_B - 60)\frac{1}{2} = (v_B + 60)\frac{1}{6}$$

$$3v_B - 180 = v_B + 60$$

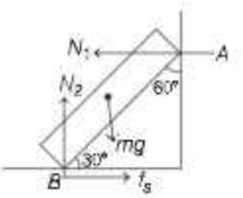
$$2v_B = 240 \Rightarrow v_B = 120 \text{ km/h}$$

Distance $= (v_B - v_S)t_1$

$$d = (120 - 60)\frac{1}{2} = 30 \text{ km}$$

$$T = \frac{d}{v_B} = \frac{30}{120} = \frac{1}{4}\text{h} = 15 \text{ min}$$

22. *(b)*



$N_2 = mg = 200$ N

Taking torque about A:

$$f_s L\cos 60^\circ + mg\frac{L}{2}\sin 60^\circ = N_2 L \sin 60^\circ$$

$$f_s\left(\frac{1}{2}\right) + 200\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = 200\frac{\sqrt{3}}{2}$$

$$f_s = 100\sqrt{3}\text{ N}$$

23. *(b)* $\omega = \sqrt{\dfrac{k}{m}} \Rightarrow \omega \propto \dfrac{1}{\sqrt{m}}$

$\because m$ decreases, so ω increases



$$\because E = \frac{1}{2}kA^2$$

The part of sand being removed, takes away a part of energy.

∴ Total energy decreases.
∴ Energy of SHM decreases.
As a result, amplitude also decreases.



24. *(c)* $T \rightarrow$ Time $[T^1]$

$S \rightarrow$ Surface Tension

$$S = \frac{F}{l} = \frac{[M^1L^1T^{-2}]}{[L^1]} = [M^1T^{-2}]$$

$A \rightarrow$ Area $[L^2]$, $\rho \rightarrow$ density $[M^1L^{-3}]$, $R \rightarrow$ radius $[L^1]$

With dimensional analysis →

$$T \propto S^{\alpha} A^{\beta} \rho^{\gamma} R^{\delta}$$

$$[M^0L^0T^1] = [MT^{-2}]^{\alpha}[L^2]^{\beta}[ML^{-3}]^{\gamma}[L]^{\delta}$$

$$[M^0L^0T^1] = [M^{\alpha+\gamma} L^{2\beta+\delta-3\gamma} T^{-2\alpha}] \quad ...(ii)$$

$$\Rightarrow -2\alpha = 1 \qquad \alpha + \gamma = 0$$

$$\Rightarrow \alpha = \frac{-1}{2} \qquad \gamma = -\alpha = \frac{1}{2}$$

$$2\beta + \delta - 3\gamma = 0 \Rightarrow 2\beta + \delta - 3\left(\frac{1}{2}\right) = 0$$

$$2\beta + \delta = 3/2$$

The best match is option (c).

25. *(c)* 10 VSD = 9 MSD

1 MSD = 0.1 cm (Given)

1 VSD = 0.9 MSD

1 VSD = 0.9 × 0.1 cm

1 VSD = 0.09 cm

L.C = 1 M.S.D − 1 V.S.D

= (0.1 − 0.09) cm = 0.01 cm

Reading = 5.08 cm

Positive zero error = 0.1 cm (Given)

∴ Final corrected reading

= (5.08 − 0.1) cm = 4.98 cm

26. *(c)* $I_d = \varepsilon_0 \frac{d\phi_E}{dt} = \varepsilon_0 \frac{d}{dt}(EA\cos 0°)$

$$I_d = \varepsilon_0 A \frac{dE}{dt}$$

$$\because \quad E = \frac{\sigma}{\varepsilon_0} \; ; I_d = \varepsilon_0 A \frac{d}{dt}\left(\frac{\sigma}{\varepsilon_0}\right)$$

$$I_d = A\left(\frac{d\sigma}{dt}\right) = \text{constant}$$

Also, B due to I_d is maximum at surface of imaginary cylinder.

27. *(a)* When unpolarised light is incident on a medium of refractive index 1.73 at Brewster's angle then reflected light is completely polarised.

Using Brewster's law, $\mu = \tan(i_p)$

$$1.73 = \tan(i_p)$$

$$i_p = 60°$$

By law of reflection, $i_p = r = 60°$

28. *(d)*



After touching charge at A,

Total charge $= q + 0 = q$

$$Q'_A = \frac{q}{2}, \quad Q'_C = \frac{q}{2}$$

After touching Charge at B,

Total charge $= q + \frac{q}{2} = \frac{3q}{2}$

$$Q'_B = \frac{3}{4}q, \quad Q'_C = \frac{3}{4}q$$

$$F = \frac{kQ^2}{r^2} \text{ (Force before touchig)}$$

Force after touching, $F' = \dfrac{k\left(\frac{q}{2}\right)\left(\frac{3q}{4}\right)}{r^2}$

$$F' = \frac{3F}{8}$$

29. *(b)* By energy conservation

$$E_1 + E_2 = E_{mix}$$

$$\frac{n_1 f_1 RT_1}{2} + \frac{n_2 f_2 RT_2}{2} = \frac{(n_1 + n_2) f RT_{mix}}{2}$$

$$\Rightarrow \frac{3}{2}p_1V_1 + \frac{3}{2}p_2V_2 = \frac{3}{2}p_{mix}(V_1 + V_2)$$

$$\frac{3}{2} \times 1 \times 2 + \frac{3}{2} \times 2 \times 3 = \frac{3}{2} \times p_{mix}(5)$$

$$\therefore \quad p_{mix} = 8/5$$

$$p_{mix} = 1.6 \text{ atm}$$

30. *(c)* $\frac{mv^2}{r} = F \qquad ...(i)$

$$mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr} \qquad ...(ii)$$

From Eq. (i)

$$\frac{m}{r}\left(\frac{nh}{2\pi mr}\right)^2 = F$$

$$r^3 = \frac{n^2h^2}{4\pi^2 mF} \Rightarrow r^3 \propto n^2 \Rightarrow r \propto n^{\frac{2}{3}}$$

From Eq. (ii)

$$v \propto \frac{n}{r} \propto \frac{n}{n^{\frac{2}{3}}} \Rightarrow v \propto n^{\frac{1}{3}}$$

31. *(a)* $\frac{T_{mercury}}{T_{earth}} = \left(\frac{r_{mercury}}{r_{earth}}\right)^{\frac{3}{2}} = \left(\frac{1}{4}\right)^{\frac{3}{2}} = \frac{1}{8}$

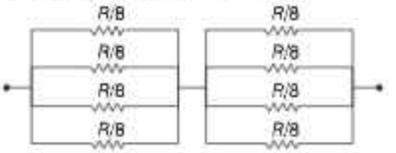
$$T_{mercury} = \frac{T_{earth}}{8} = \frac{687}{8} = 85.875 \text{ days}$$

$$= 86 \text{ earth days}$$

32. *(b)* $w' = mg' = mg\left(\frac{R}{R+h}\right)^2$

$$= 48\left(\frac{R}{R + \frac{R}{3}}\right)^2 = 48 \times \frac{9}{16} = 27\text{N}$$

33. *(c)* Resistance of each part = $R/8$



$$R_{eq} = \frac{1}{4}\left(\frac{R}{8}\right) + \frac{1}{4}\left(\frac{R}{8}\right) = \frac{R}{16}$$

34. *(b)* Given, $n = 2, \; Z = 1$

$$2\pi r = n\lambda \Rightarrow 2\pi\left(0.052 \times \frac{n^2}{Z}\right) = n\lambda$$

On solving, $\lambda = 0.65$ Nm

35. *(b)* Given, $p = 5 \times 10^{-9}$ Cm,

$E = 4 \times 10^5$ N/C, $\theta_i = 0°$, $\theta_f = 60°$

$$\Delta U = U_f - U_i$$

$$= -pE\cos\theta_f - [-pE\cos\theta_i]$$

$$= pE(\cos\theta_i - \cos\theta_f)$$

$$= 5 \times 10^{-9} \times 4 \times 10^5\left(1 - \frac{1}{2}\right) = 1 \text{ J}$$

36. *(b)* Resistance across AC,

$$R_{AC} = \frac{3 \times 1}{3 + 1} = \frac{3}{4}\Omega$$

Resistance across CB, $R_{CB} = \frac{2 \times 4}{2 + 4} = \frac{4}{3}\Omega$

Total resistance, $R_{AB} = \frac{3}{4} + \frac{4}{3} = \frac{25}{12}\Omega$

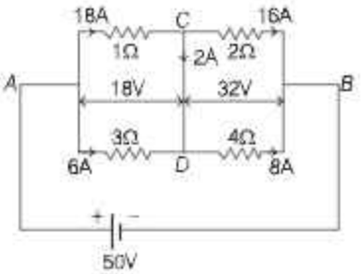
Total current, $I_{AB} = \frac{50 \times 12}{25} = 24$ A

Voltage drop, $V_{AC} = 24 \times \frac{3}{4} = 18$ V

Voltage drop, $V_{CB} = 24 \times \frac{4}{3} = 32$ V

Now, $\quad I_{AC} = \frac{18}{1} = 18$ A

$\Rightarrow \quad I_{CB} = \frac{32}{2} = 16$ A



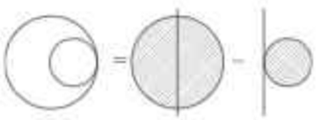
By KCL, current in the branch $CD = 18 - 16 = 2$A

37. *(c)* Photon and electron have same energy E.

$$\frac{\lambda_{photon}}{\lambda_{electron}} = \frac{hc/E}{\frac{h}{\sqrt{2mE}}} = c\sqrt{\frac{2m}{E}}$$

38. *(a)* $I \propto$ intensity, so curve A is correct.
Photo current does not depend linearly on frequency.
So, graph C and D are not correct.

39. *(c)*



$$I_1 = \frac{2}{5}M(2R)^2, \quad I_2 = \frac{2}{5}mr^2 + mr^2$$

$$I_2 = \frac{2}{5}\left(\frac{M}{8}\right)R^2 + \frac{M}{8}R^2 = \frac{7}{40}MR^2$$

$$I = I_1 - I_2 = \frac{8MR^2}{5} - \frac{7MR^2}{40} = \frac{57}{40}MR^2$$

$$\frac{I_2}{I} = \frac{7}{57}$$

40. *(b)* At $t = 15$ ms

$$V_{in} = 220\sin(100\pi \times 15 \times 10^{-3})$$

$$= 220\sin\left(\frac{3\pi}{2}\right) = -220\text{V}$$

So, diode D_1 is in reverse bias and D_2 is in forward bias.

41. *(b)* According to question
Pressure is constant, $p_A = p_B$, $Q_A = Q_B$

$$\Delta U_A = \Delta U_B$$

According to 1st law of thermodynamics

$$Q = \Delta U + W \Rightarrow W_1 = W_2$$

$$p_1\Delta V_1 = p_2\Delta V_2 \Rightarrow (\pi r_A^2)X_A = (\pi r_B^2)X_B$$

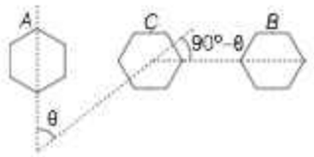
$$\frac{r_A}{r_B} = \sqrt{\frac{X_B}{X_A}} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

42. *(c)* $P = a^3b^2c^{-1}d^{\frac{-1}{2}}$

$$\left(\frac{\Delta P}{P} \times 100\right) = \left(\frac{3\Delta a}{a} + \frac{2\Delta b}{b} + \frac{1\Delta c}{c} + \frac{1}{2}\frac{\Delta d}{d}\right) \times 100$$

$$= 3(1\%) + 2(3\%) + 1(2\%) + \frac{1}{2}(4\%) = 13\%$$

43. *(c)* Initially two polaroids are crossed, i.e. 90°



After passing from first polaroid intensity is I_0

I unpolarsed

$$\xrightarrow{A} I_0 \xrightarrow{C} I_0\cos^2\theta \xrightarrow{B}$$

$$I_0\cos^2\theta\cos^2(90^\circ - \theta)$$

$$= \frac{I_0}{4}\sin^2(2\theta) = \frac{I_0}{4}\sin^2(2 \times 22.5) = \frac{I_0}{8}$$

44. *(d)* Given, $m_P = m_Q$

Also, $(v_{max})_P = (v_{max})_Q$

$$\therefore A_P\omega_P = A_Q\omega_Q \Rightarrow A_P\sqrt{\frac{k_1}{m_P}} = A_Q\sqrt{\frac{k_2}{m_Q}}$$

$$\left[\because \omega = \sqrt{\frac{k}{m}}\right]$$

$$\therefore \quad \frac{A_Q}{A_P} = \sqrt{\frac{k_1}{k_2}}$$

45. *(b)* For open pipe,

$$f = \frac{v}{2l}$$

When half of its length is dipped in water, Now it is closed open pipe

$$f' = \frac{v}{4\left(\frac{l}{2}\right)}$$

$$f' = \frac{v}{2l} \Rightarrow f' = f$$

Chemistry

46. *(d)* For $2 \rightarrow 3$ transition $n_1 = 2$

$$n_2 = 3$$

For $4 \rightarrow 6$ transition,

$$n_1 = 4$$

$$n_2 = 6$$

Using, $\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

On substituting values, we get

For $2 \rightarrow 3$ transition $\frac{1}{\lambda_1} = R(1)^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$ ...(i)

For $4 \rightarrow 6$ transition $\frac{1}{\lambda_2} = R(1)^2\left[\frac{1}{4^2} - \frac{1}{6^2}\right]$ ...(ii)

Divide Eq. (i) by Eq. (ii),

$$\frac{\lambda_2}{\lambda_1} = \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{4^2} - \frac{1}{6^2}}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{4}{1}$$

$$\Rightarrow \quad \frac{\lambda_1}{\lambda_2} = \frac{1}{4}$$

47. *(b)* Statements given in C and E are correct. While A, B and D are incorrect. Their correct form is

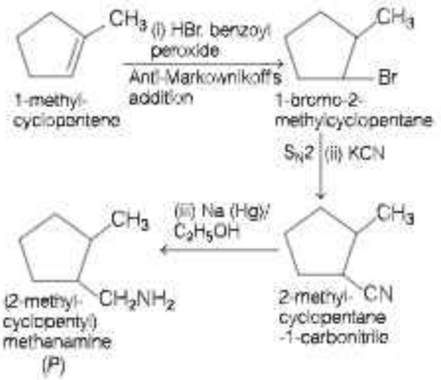
(A) Both gallium (Ga) and caesium (Cs) have relatively low melting points.

(B) On pauling scale, the electronegativity values of N and Cl are same.

(D) The correct order of the first I.E. of given elements is Na < Al < Mg < Si.

48. *(b)* The correct match is
A-III, B-IV, C-I, D-II.

49. *(a)* The complete reaction sequence is as follows.



50. *(a)* For calculating energy, use the formula, $E_n = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2}$ J

For He^+, $Z = 2$, $n = 1$

On substituting values, we get

$$E_1 = -2.18 \times 10^{-18} \times \frac{(2)^2}{(1)^2}\text{ J}$$

$$E_1 = -8.72 \times 10^{-18}\text{ J}$$

For Li^{2+}, $Z = 3$, $n = 1$

On substituting values, we get

$$E_1 = -2.18 \times 10^{-18} \times \frac{(3)^2}{(1)^2}\text{ J}$$

$$= -19.62 \times 10^{-18}\text{ J}$$

Now, to calculate radius, we use the formula, $r_n = 52.9 \times \frac{n^2}{Z}$ pm

For He^+, $Z = 2$, $n = 1$

On substituting values, we get

$$r_1 = 52.9 \times \frac{1^2}{2} = 26.4\text{ pm}$$

For Li^{2+}, $Z = 3$, $n = 1$

$$r_1 = 52.9 \times \frac{1^2}{3} = 17.6\text{ pm}$$

Thus, option (a) is the correct answer.

51. *(c)* Among the given complexes, $[NiCl_4]^{2-}$ and $[Ni(H_2O)_6]^{2+}$ has 2 unpaired electrons. Thus, these are paramagnetic in nature.

(A) In $[NiCl_4]^{2-}$, Ni is in + 2 oxidation state. Thus, its electronic configuration [Ar] $3d^8 4s^0$. Also Cl^- is a weak field ligand. So, no pairing will take place.

$Ni^{2+} =$ ↑↓ | ↑↓ | ↑↓ | ↑ | ↑ → 2 unpaired electrons

(D) In $[Ni(H_2O)_6]^{2+}$, Ni is in + 2 oxidation, thus, its electronic configuration $[Ar]3d^8 4s^0$.

Also, H_2O is a weak field ligand. So, no pairing will take place.

Ni^{2+} = ↑↓ ↑↓ ↑↓ ↑ ↑ → 2 unpaired electrons

52. *(c)* Statement I is correct but statement II is incorrect. It is because, all the elements of N-family form two type of oxides i.e., E_2O_3 and E_2O_5.

53. *(b)* Elements of *s*-block and *p*-block are considered as main group elements. So, the element with electronic configuration [Ne] $3s^1$ i.e sodium and [Kr] $4d^{10}5s^25p^5$ i.e., iodine are main group elements. So, option (b) is the correct answer.

54. *(d)* Dalton's atomic theory successfully explained law of conservation of mass, law of constant proportion and law of multiple proportion. However it failed to explained law of gaseous volume.

55. *(a)* In KO_2, K is in + 1 oxidation state. It is because alkali metal always show + 1 oxidation state in its compounds.

- For H_2O, Let oxidation number of O be x $2(+1) + 2x = 0 \Rightarrow x = -1$

  Let oxidation number of S be x.
- For H_2SO_4, $2(+1) + x + 4(-2) = 0$ $\Rightarrow x = +6$

Thus, the oxidation number of underlined elements respectively is +1, −1 and + 6.

56. *(d)* Given, $t_{1/2} = 1$ min

$t_{99.9\%} = ?$

For first order reaction $t_{99.9\%}$ is given by

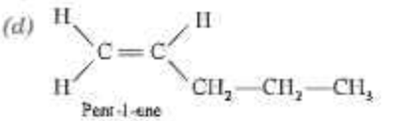
$t_{99.9\%} = 10 \times t_{1/2} = 10 \times 1 = 10$ minutes

57. *(b)* As the strength of ligand increases, Δ_o and absorbed energy increases and absorbed wavelength decreases. So, the correct order of wavelength of light absorbed by given complexes is

$[Co(CN)_6]^{3-} < [Co(NH_3)_6]^{3+} < [Ti(H_2O)_6]^{3+} < [Cu(H_2O)_4]^{2+}$

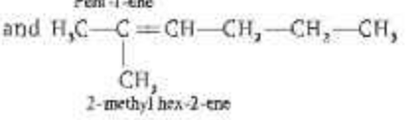
i.e., B < A < D < C

58. *(d)*



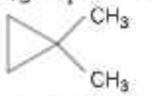
Pent-1-ene

and



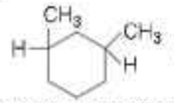
2-methyl hex-2-ene

will not show geometrical or *cis-trans* isomerism as each terminal individually does not have different atom/group attached to it.

-  will not show

1,1-dimethylcyclopropane

geometrical or *cis-trans* isomerism as two sp^3 C of ring individually does not have different atoms/groups attached to it.

-  will show *cis-trans*

1,2-dimethylcyclohexane

or geometrical isomerism. It is because two sp^3 C of ring individually have different atom/group of atom attached to it.

59. *(d)* Ionisation of phosphoric acid occurs as below—

$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-, K_{a_1}$

$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}, K_{a_2}$

$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}, K_{a_3}$

$H_3PO_4 \rightleftharpoons 3H^+ + PO_4^{3-}$

$\Rightarrow K = K_{a_1} \times K_{a_2} \times K_{a_3}$ ...(i)

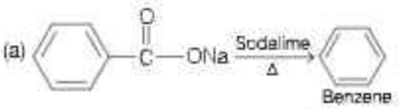
Now, as negative charge increases, tendency to loose H^+ ions decreases in polybasic acid. So, the order of ionisation constant is $K_{a_1} > K_{a_2} > K_{a_3}$.

Also taking log of eq. (i) we get,

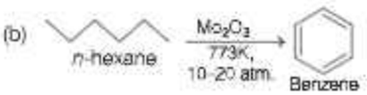
$\log K = \log K_{a_1} + \log K_{a_2} + \log K_{a_3}$.

Thus, statements given in A, B and C are correct while D is incorrect.

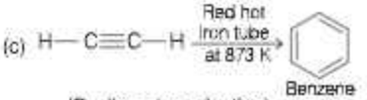
60. *(d)* The complete reaction for the given options are as follows.

(a) 

Benzene

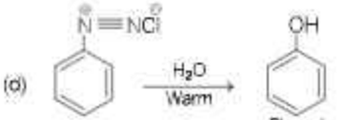
(b) 

Benzene

(c) 

(Cyclic polymerisation)

Benzene

(d) 

Phenol

Thus, reaction given in option (d) will not give benzene.

61. *(c)* Given, $\Lambda_+^\circ = 349.6$ S cm^2 mol^{-1}

$\Lambda_-^\circ = 50.4$ S cm^2 mol^{-1}

$\Lambda_m = 90$ S cm^2 mol^{-1}

$\alpha = ?$

Using formula

$\Lambda_m^\circ = \Lambda_+^\circ + \Lambda_-^\circ$

$= 349.6 + 50.4$

$= 400$ S cm^2 mol^{-1}

$\Rightarrow \quad \alpha = \dfrac{\Lambda_m}{\Lambda_m^\circ} = \dfrac{90}{400} = 0.225$

62. *(b)* Both Statement I and Statement II are false. Their correct form are

- a hypothetical diatomic molecule with zero bond order does not exist for example He_2, Be_2 etc.
- as bond order increases bond length decreases.

63. *(a)* Conductivity depends upon the number of ions produced when a compound is dissolved. The more the number of ions, more will be the conductivity. Among the given options, $[Co(NH_3)_4Cl_2]$, $[Co(NH_3)_6]Cl_3$ and $[Co(NH_3)_5Cl]Cl$ will give ions when dissolved. But $[Co(NH_3)_3Cl_3]$ does not dissociate into ions in solution, rather exists as neutral complex. Thus, it will show minimum conductance.

64. *(a)* The correct match is A-II, B-I, C-IV, D-III.

65. *(a)* For the given reaction,

$C(s) + 2H_2(g) \longrightarrow CH_4(g)$

Since, ΔH is negative, thus the reaction is exothermic i.e. $H_R > H_P$.

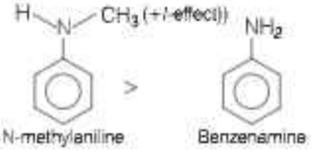
So, option (a) is the correct answer.

66. *(b)* The correct match is A-II, B-I, C-IV, D-III,

67. *(c)* The correct decreasing order of basicity is

$(C_2H_5)_2NH$ > $C_2H_5NH_2$ >

N-ethylethanamine Ethanamine



N-methylaniline Benzenamine

68. *(b)* To calculate the number of atoms, use the formula,

No. of atoms = Atomicity × Moles × N_A

(A) In 212 g of $Na_2CO_3(s)$

No. of atoms $= 6 \times \dfrac{212}{106} \times N_A = 12N_A$

(B) In 248 g of $Na_2O(s)$

No. of atoms $= \frac{3 \times 248 \times N_A}{62} = 12 N_A$

(C) In 240 g of NaOH(s)

No. of atoms $= \frac{3 \times 240 \times N_A}{40} = 18 N_A$

(D) In 12 g of H_2

No. of atoms $= \frac{2 \times 12 \times N_A}{2} = 12 N_A$

(E) In 220 g of $CO_2(g)$

No. of atoms $= \frac{3 \times 220 \times N_A}{44} = 15 N_A$

So, A, B and C have equal number of atoms.

69. *(b)* The correct match is A-IV, B-III, C-I, D-II.

70. *(c)* Acidic strength increases with $-I$ effect while it decreases with $+I$ effect. So, the correct order is

$HCOOH > CH_3COOH > (CH_3)_2CHCOOH > (CH_3)_3C(COOH)$.

71. *(c)* Statement I is true but Statement II is false. The correct form of statement II is

The number of unpaired electrons in Cr^{2+} ion is 4 while in Nd^{3+} it is 3.

Thus, they are not equal.

72. *(a)* The correct match is A-IV, B-III, C-I, D-II.

73. *(c)* Given, $K_a = 2500\, K_f$

$T = 1000$ K

$R = 0.0831$ L atm mol^{-1} K^{-1}

$K_p = ?$

Using, $K_c = \frac{K_f}{K_a}$

$= \frac{K_f}{2500\, K_f} = \frac{1}{2500}$

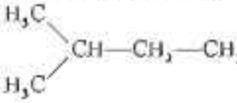
Now, use the formula.

$K_p = K_c (RT)^{\Delta n_g}$

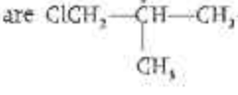
$= \frac{1}{2500}(0.0831 \times 1000)^1 = 0.033$

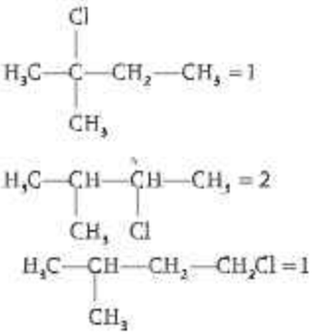
74. *(a)* Both Statement I and Statement II are correct.

75. *(d)* The products obtained on monochlorination of



are

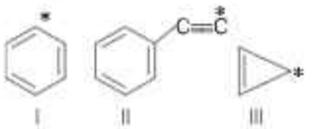




Where, * represent chiral carbon.

So, total 6 products including stereoisomers are expected.

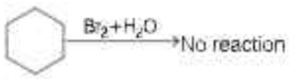
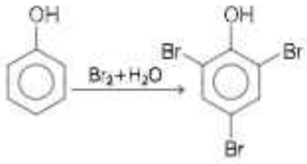
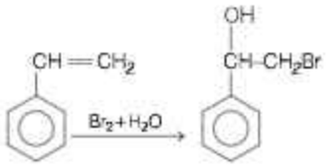
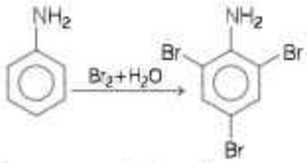
76. *(a)* As the stability of free radical (obtained after C—H bond fission homolytically) increases, the bond dissociation energy decreases. Thus, for the radicals,



The order of stability is III > I > II.

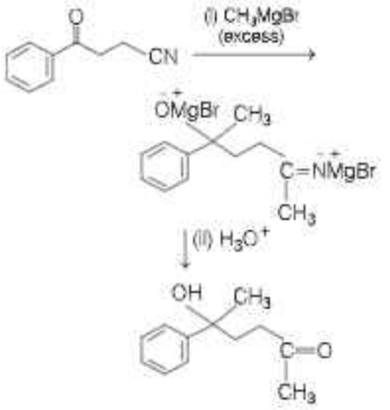
So, the order of B.D.E is II > I > III.

77. *(a)*

- 
- 
- 
- 

∴ Bromine water solution is decolourised in reaction (b), (c) and (d) but not in (a).

78. *(b)* The complete reaction sequence is as follows



79. *(c)* Using formula, $\Delta T_b = iK_b m$

Let $m = M$ and K_b is same for all more the value of iM more will be the value of ΔT_b and more will be the boiling point.

- For 0.01 M urea $iM = 1 \times 0.01 = 0.01$
- For 0.01 M KNO_3, $iM = 2 \times 0.01 = 0.02$
- For 0.01 M Na_2SO_4, $iM = 3 \times 0.01 = 0.03$
- For 0.015 M $C_6H_{12}O_6$ $iM = 1 \times 0.015 = 0.015$

Thus, 0.01 M Na_2SO_4 will exhibit highest boiling point.

80. *(c)* The correct match is A-I, B-II, C-III, D-IV.

81. *(b)* Given, $n_x = 5$

$n_y = 10$

$\Rightarrow$ $p_x^\circ = 63$ torr

$p_y^\circ = 78$ torr

Using formula, $p_{S(calc.)} = p_x^\circ \cdot n_x + p_y^\circ \cdot n_y$

$p_{S(calc.)} = \frac{63 \times 5}{15} + 78 \times \frac{10}{15}$

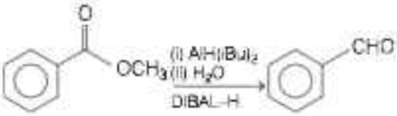
$p_{S(cal.)} = 21 + 52 = 73$ torr

$\because p_{S(obs)} < p_{S(calc.)}$

so, it will show negative deviation.

82. *(b)* The given characteristics are of D-fructose. It is a keto sugar that exists in α and β anomeric forms. It is laevorotatory, that is found in honey.

83. *(b)* The suitable reagent for conversion of ester to aldehyde is $AlH(iBu)_2$ or DIBAL—H_2.



84. *(a)* Both A and R are true and R is the correct explanation of A.

85. *(a)* Given, $\Delta H_f^\circ(SO_4^{2-}) = -216$ kcal/mol

$\Delta H_f^\circ(BaSO_4) = -349$ kcal/mol

$\Delta H_f^\circ(Ba^{2+}) = ?$

$\Delta H_{rxn}^\circ(BaSO_4) = -4.5$ kcal mol^{-1}

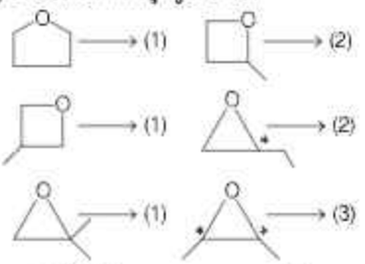
$\Delta H_{rxn}^\circ(BaSO_4) = \Delta H_f^\circ(BaSO_4)$

$-[\Delta H_f^\circ(Ba^{2+}) + \Delta H_f^\circ(SO_4^{2-})]$

$-4.5 = [-349] - [\Delta H_f^\circ(Ba^{2+}) + (-216)]$

$\Delta H_f^\circ(Ba^{2+}) = -128.5$ kcal mol^{-1}

86. *(c)* Total number of possible isomer of cyclic ether of C_4H_8O is 10.



So, total 10 isomers are possible.

87. *(a)* Properties given in A and D are correct while B and C are incorrect. The correct order of B and C are

(B) $XeF_2 > XeF_4 > XeO_3$. (Number of lone pair on central atom)

(C) O—H < C—H < N—O Bond length

88. *(d)* For the reaction,

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, the higher yield of NO is obtained by higher temperature, higher concentration of N_2 and higher concentration of O_2.

Thus, option (d) is the correct answer.

89. *(a)* Given, $k = 0.03\ s^{-1}$

$[A_0] = 7.2$ mol L^{-1}; $[A_t] = 0.9$ mol L^{-1}

$\log 2 = 0.3010$

From the unit of k, it is clear that, reaction is of first order.

Using formula, $k = \frac{1}{t} 2.303 \log \frac{[A_0]}{[A_t]}$

On substituting values, we get

$t = \frac{1}{0.03} \times 2.303 \log \frac{7.2}{0.9} \Rightarrow t = 69.3$ seconds

90. *(d)* Reaction given in option (d) i.e.,

$2CuO + C \xrightarrow{\Delta} 2Cu + CO_2$ is

used to detect carbon. While Lassaigne's test is used to detect phosphorus, sulphur, nitrogen and halogens. Thus, reaction given in (d) does not belongs to Lassaigne's test.

Biology

91. *(b)* Complex-II of the mitochondrial electron transport chain is also known as succinate dehydrogenase. It links the citric acid cycle and ETC by transferring electrons from succinate to ubiquinone. Unlike other complexes, it doesn't pump protons but still contributes to ATP production via $FADH_2$ oxidation.

92. *(b)* PCR amplifies DNA exponentially. After each cycle, DNA doubles, so the total number of DNA copies after n cycles is 2^n. This exponential growth enables small amounts of DNA to be analysed or used for diagnostics and research purposes.

93. *(a)* Drawbacks of the IVF are given in statements B, D and F. These can be explained as

In Vitro fertilisation (IVF) is an assisted reproductive technology where egg is fertilised by sperm outside the body in a lab, and then the resulting embryo is transferred to the uterus of the mother.

This is a technologically advanced procedure requiring specialised equipment, laboratory facilities, and expensive medications (reagents). This makes it a costly treatment. So, this is a potential drawback.

Also, due to the possibility of having ones own child through IVR, the adoption of orphans has seen a decline in number. Moreover,the early embryo survival rates are lower in IVF compared to those in natural conception.

94. *(d)* In frogs, the vena cava is the major vein that returns deoxygenated blood from various parts of the body to sinus venosus which empties the blood into the right atrium, allowing it to be pumped towards the lungs and skin for oxygenation. It is a crucial step for the amphibian's incomplete double circulatory system.

95. *(a)* Reductionist biology aims to understand complex biological systems by breaking them down into simpler physical and chemical components. It uses physics and chemistry to explain biological processes at the molecular and cellular levels, forming the foundation of molecular biology, biochemistry, and biophysics.

96. *(a)* Both statement I and statement II are correct.

RNA likely evolved first with dual roles—genetic and catalytic—but is unstable. DNA evolved later from RNA. It provides more stability and repair mechanisms, better supporting long-term information storage. Both statements align with the widely accepted RNA world hypothesis.

97. *(a)* Epiphytes grow on mango branches to access sunlight and air, but do not harm or benefit the mango tree. This interaction is commensalism, where one organism benefits, while the other is unaffected. The mango provides support, without any nutritional exchange or competition.

98. *(a)* Statements A, B and C are true, while statements D and E are incorrect. These can be corrected as

In eukaryotes, ribosomes are 80S, consisting of a 60S large subunit and a 40S small subunit.

In prokaryotes, ribosomes are 70S, made up of a 50S large subunit and a 30S small subunit.

99. *(c)* Ex-situ conservation refers to the conservation of species outside their natural habitats. Zoos and botanical gardens are prime examples, where plants and animals are protected and breed in controlled environments. This contrasts with *insitu* conservation (e.g., national parks, wildlife sanctuaries, etc), where species are protected in their natural habitats.

100. *(c)* Statement I is correct, but statement II is incorrect. The incorrect statement can be corrected as

The rate of production of organic matter during photosynthesis in an ecosystem is often referred to as Gross Primary Productivity (GPP). However, Net Primary Productivity (NPP) is the stored plants produce organic matter available for consumption by herbivores and it accounts for the energy used by plants for their own metabolic processes (respiration).

101. *(d)* The correct match is A-II, B-III, C-IV, D-I.

Emphysema involves damaged alveolar walls and a decreased respiratory surface, leading to breathing difficulties.

Angina pectoris is acute chest pain caused when the heart doesn't get enough oxygen.

Glomerulo-nephritis is an inflammation of the glomeruli in the kidneys, affecting filtration.

Tetany is a condition characterised by rapid spasms in muscles, often due to low calcium levels in the body.

102. *(b)* Both A and R are true, but R is not the correct explanation of A.

The lack of colour and nectar in both wind and water pollinated flowers is

The lack of colour and nectar in both wind and water pollinated flowers is because these flowers do not need to attract animal pollinators.

The production of large amount of pollens is a mechanism ensuring successful pollination.

103. *(c)* Beer is a non-distilled alcoholic beverage made by the fermentation of sugars by yeast, whereas, whisky, brandy and rum are distilled alcoholic beverages.

104. *(d)* Statement I is incorrect, but statement II is correct. The incorrect statement can be corrected as

In floral formulas:

⊖ indicates actinomorphic (radial symmetry) flowers, not zygomorphic (bilateral symmetry).

G represents female reproductive whorl, gynoecium (ovary). An inferior ovary would be represented by $\overline{G}$.

A bar below G i.e., $\underline{G}$, represents a superior ovary.

105. *(d)* Streptokinase, produced by the bacterium *Streptococcus* and modified by genetic engineering, is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to the heart attack.

106. *(c)* Chromosome-1 in the human genome has the highest number of genes 2968, which is crucial for many essential processes in the body.

107. *(d)* In male frogs, copulatory pads are located on the first digit of the forelimbs. These pads help the male frog grasp the female during mating, ensuring that the male can hold onto the female securely, while she lays her eggs.

108. *(d)* Cytokinins promote nutrient mobilisation and delay leaf senescence by preserving chlorophyll and supporting protein synthesis.

They help redirect nutrients to active, growing parts of the plant, keeping older leaves green for longer period of time. In contrast, hormones like ethylene and abscisic acid generally promote aging or stress responses in plants.

109. *(b)* A pseudocoelomate has a body cavity not fully lined by mesoderm. In this case, mesodermal tissue is present only along the body wall but absent around the alimentary canal, indicating a pseudocoelom.

True coelomates (schizocoelomates) have mesoderm on both sides, while acoelomates have no body cavity at all.

110. *(a)* The correct match is A-IV, B-III, C-I, D-II.

The head of the sperm contains the nucleus that contains genetic material (DNA).

The middle piece contains mitochondria, which produce ATP for movement of sperm.

The acrosome contains enzymes like hyaluronidase that help penetrate the egg.

The tail (flagellum) enables the sperm to swim.

111. *(a)* Correct sequence of stages in life cycle of Pteridophytes stages in the B, A, D, E, C.

The diploid sporophyte of the pteridophyte undergoes meiosis to form haploid spores. These spores germinate into the gametophyte, known as the prothallus. The prothallus develops male and female sex organs also called antheridia and archegonia respectively. Male gametes or sperms swim through water to reach the egg. The male and female gametes fuse to form a diploid zygote, restarting the sporophyte generation.

This represents the alternation of generations in pteridophytes: from diploid sporophyte → haploid spores → haploid gametophyte → gametes → zygote → sporophyte.

112. *(a)* A, B and C can regulate the cardiac activities of the heart.

The heart's intrinsic rhythm is regulated by nodal tissues like the Sino Atrial (SA) and Atrio Ventricular (AV) nodes.

A special neural centre in the medulla oblongata controls the rate of heartbeat via autonomic nerves.

Adrenal medullary hormones like adrenaline and noradrenaline increase heart rate and force of contraction.

On the other hand, adrenal cortical hormones mainly regulate metabolism, electrolyte balance, and immune response. They are not directly involved in cardiac activity.

113. *(b)* *Volvox* is a green alga (a colonial freshwater organism) that performs photosynthesis. However, it lacks specialised cells like heterocysts, which are needed for nitrogen fixation. The latter is typically carried out by certain cyanobacteria (like *Anabaena*, *Nostoc* and *Oscillatoria*) or specific bacteria (like *Azotobacter*).

114. *(d)* Statement I is incorrect, but statement II is correct. The incorrect statement can be corrected as

Transfer RNAs (*t*RNAs) and ribosomal RNAs (*r*RNAs) are crucial for mRNA translation. *t*RNAs bring specific amino acids to the ribosome based on their anticodon sequence matching the *m*RNA codon, and rRNA forms the structural backbone of the ribosome, facilitating the process of protein synthesis. Thus, they directly interact with *m*RNA.

115. *(c)* When an alien piece of DNA is inserted into a plasmid at an EcoRI site, it often disrupts a gene involved in β-galactosidase production. This is typically used in lacz gene systems (e.g., in blue/white screening). The insertion of the alien DNA disrupts the lacz gene, preventing the breakdown of X-gal, resulting in white colonies. Thus, white colonies are selected because they represent successful insertion of the foreign DNA, disrupting the lacz gene and preventing the blue colouration.

116. *(a)* Eli Lilly used genetically engineered bacteria (*Escherichia coli*) to produce human insulin. The process involved inserting the human insulin gene into a bacterium's plasmid, which then allowed the bacterium to produce insulin. This is one of the first widely known examples of recombinant DNA technology in medicine.

117. *(c)* Transferase catalyses the transfer of a group, G (other than hydrogen) between a pair of substrate S and S^* as given $S - G + S^* \rightarrow S + S^* - G$.

Example of such an enzyme is glutamate pyruvate transaminase.

118. *(a)* Statement given in option (a) is not correct. It can be corrected as

In monocot stems, the hypodermis is primarily composed of sclerenchyma cells, which provide structural support and rigidity. While collenchyma cells can also contribute to hypodermis support, especially in young stems and other parts of the plant, sclerenchyma is the dominant tissue in monocot hypodermis. Parenchyma cells, on the other hand, are typically not found in the hypodermis of monocot stems.

119. *(c)* The correct sequence of events in the life cycle of bryophytes is B, E, A, D and C.

Gametophyte of the bryophyte first attaches to a suitable surface (substratum) for growth.
It contains male gametes (antherozoids), which are released into water, where they swim toward the female gametes (eggs).
Fertilisation occurs when an antherozoid fuses with the egg, forming a zygote.
The fertilised egg develops into a sporophyte, which grows attached to the gametophyte.
The sporophyte undergoes meiosis to produce haploid spores, thus, completing the cycle.

120. *(d)* Statements B, D and E are incorrect . These can be corrected as
Chemotherapeutic drugs target cancerous cells, but they can also harm normal (non-cancerous) cells, leading to side effects. They are not used specifically to kill non-cancerous cells.
Chemotherapeutic drugs are primarily cytotoxic and work by directly killing cancer cells, while biological response modifiers (like interferons and interleukins) modulate the body's immune response.
In leukaemia, the blood cell counts can either be elevated (due to the uncontrolled production of abnormal white blood cells) or be decreased in advanced stages due to suppression of normal blood cell production.

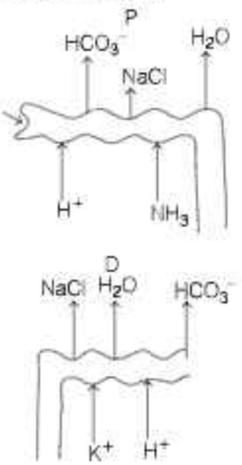
121. *(d)* The correct match is A-II, B-III, C-I, D-IV.
The centromere is essential for cell division as it anchors spindle fibers.
Cilia are hair-like structures that aid in cell movement or fluid transport.
Cristae are folds within mitochondria that increase surface area for energy production.
The cell membrane, made of protein and phospholipid bilayer, regulates materials exchange and protects the cell, maintaining its internal environment.

122. *(b)* The correct match is A-III, B-I, C-II, D-IV.
Chlorophyll *a* is the primary photosynthetic pigment and appears blue-green in colour.
Chlorophyll *b* complements chlorophyll a and is yellow-green in colour.
Xanthophyll is a type of carotenoid pigment responsible for yellow colouration.
Carotenoids are pigments that range from yellow to orange and are also involved in light absorption for photosynthesis.

123. *(d)* Statements A, C, D and E are correct. The incorrect statement is B and can be corrected as
In human pregnancy, the major organ systems are not fully formed by the end of 8 weeks. On the other hand, basic structures and organ systems begin to develop during the embryonic stage (weeks 3-8). The major organ systems are formed at the end of 12 weeks.

124. *(d)* In the seeds of cereals (such as wheat and maize), the aleurone layer is a protein-rich layer that forms the outermost part of the endosperm. It separates the endosperm from the embryo and plays a crucial role during germination by secreting enzymes like amylase, which help digest the stored starch in the endosperm.

125. *(b)* Diagram given in option (b) is correct regarding proximal and distil tubule of the nephron.



PCT helps to maintain the pH and ionic balance of the body fluids by selective secretion of hydrogen ions and ammonia into the filtrate and by absorption of HCO_3^- from it.
DCT is capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions and NH_3 to maintain the pH and sodium-potassium balance in blood.

126. *(d)* C represents foam breaker in the bio-reactor. It prevents excessive foam formation during fermentation caused by aeration and mixing. It can be mechanical (using rotating paddles) or chemical (using antifoaming agents). Foam control is essential to avoid contamination, ensure proper oxygen transfer, and maintain effective bioprocess operation.

127. *(c)* A is true, but R is false. R can be corrected as
In angiosperms, the egg apparatus consists of one egg cell and two synergids. The two polar nuclei are actually located in the central cell of the embryo sac and do not form any part of the egg apparatus.

128. *(a)* Mesosome is a specialised membranous structure found in prokaryotic cells. These structures are invaginations of the cell membrane that involved in cell wall formation, DNA replication, and respiration and also aid in cellular processes such as division and energy production.

129. *(b)* B, C, D are the post-transcriptional events in an eukaryotic cell.
Removal of introns and joining of exons is known as splicing, which removes non-coding sequences (introns) and joins the coding sequences (exons).
Addition of methyl group at 5' end of *hn*RNA is the capping process, where a methylated cap is added to the 5' end of the *hn*RNA (heterogeneous nuclear RNA) for stability and to facilitate translation.
Addition of adenine residues at 3' end of hnRNA is the polyadenylation process, where a poly-A tail is added to the 3' end of the hnRNA for stability and regulation.

130. *(b)* The pattern of inheritance for polygenic traits is non-mendelian.
These traits are influenced by multiple genes and their expression can vary due to the combined effects of these genes and environmental factors.
Unlike Mendelian traits, polygenic traits do not follow simple dominant or recessive inheritance patterns.

131. *(d)* Catalase is an enzyme that contains haem as its prosthetic group. Haem is an iron-containing molecule that plays a crucial role in the enzyme's ability to catalyse the breakdown of hydrogen peroxide into water and oxygen. This is an important function for protecting cells from oxidative damage.

132. *(b)* Organisms arranged in increasing order of complexity of body organisation will be in sequence C,E,A,D,B.
Prokaryotes like bacteria with cell wall made of polysaccharides and amino acids, which are the simplest

organisms with no membrane-bound organelles(C).
Unicellular eukaryotes (e.g., protists) with cellular body organisation, where the organism is composed of a single cell, but it still has a complex internal structure compared to prokaryotes(E).
Fungi are multicellular heterotrophs with cell wall made of chitin, which are multicellular eukaryotes with a chitin-based cell wall. They exhibit a higher level of organisation than unicellular organisms(A).
Plants are eukaryotic autotrophs with tissue/organ level of body organisation , which are eukaryotic organisms that have tissue and organ systems (e.g., roots, stems, leaves) and are autotrophic (can produce their own food via photosynthesis)(D).
Animals are heterotrophs with tissue/organ/organ system level of body organisation, which are eukaryotic, multicellular, heterotrophic organisms with complex body structures (tissues, organs, organ systems)(B).

133. *(b)* Ramdeo Misra is known as the "father of ecology in India" for his significant contributions to the field of ecology and environmental studies in the country.

134. *(c)*The correct match is A-IV, B-III, C-I, D-II.
Alfred Hershey and Martha Chase confirmed DNA as the genetic material using their famous experiment with bacteriophages, which demonstrated that DNA is responsible for carrying genetic information.
Frederick Griffith is famous for his experiment with *Streptococcus pneumoniae*, showing that genetic material could be transferred between bacteria, a process known as transformation.
Euchromatin and heterochromatin are associated with the packing of chromatin in the cell, with euchromatin being loosely packed and light-stained, whereas heterochromatin being densely packed and dark-stained.

135. *(b)* Neoplastic characteristics refer to the abnormal growth and proliferation of cells, which leads to the formation of a neoplasm or tumour. The key features include:
A mass of proliferating cells (neoplastic cells proliferate abnormally).
Rapid growth of cells (neoplastic cells grow uncontrollably).
Invasion and damage to the surrounding tissue (neoplastic cells can invade nearby tissues and spread).

136. *(a)* Both statement I and statement II are correct.
DNA fragments obtained from gel electrophoresis can be used in constructing recombinant DNA by isolating specific fragments for further manipulation.
In gel electrophoresis, smaller DNA fragments move faster through the gel and travel farther towards the anode (positive end). Larger DNA fragments move more slowly and remain closer to the wells (negative end). Therefore, smaller DNA fragments are observed near the anode and larger fragments near the wells.

137. *(c)*The correct match is A-III, B-II, C-I, D-IV.
Adenosine is a nucleoside, as it consists of a nitrogen base (adenine) and a sugar.
Adenylic acid is a nucleotide, containing a phosphate group.
Adenine is a nitrogen base, a building block of nucleic acids.
Alanine is an amino acid, a basic unit of proteins.

138. *(a)* A and B are true, whereas C and D are false. These incorrect statements can be corrected as
The first polar body is produced after the first meiotic division and is associated with the formation of secondary oocyte.
The Luteinizing Hormone (LH) surge causes ovulation, not the disintegration of the endometrium. Menstrual bleeding occurs due to a drop in progesterone when the fertilised egg does not fertilise.

139. *(d)* All living members of the class.
Cyclostomata, which includes species like lamprey and hagfish, are ectoparasites. They attach themselves to the bodies of other fish and feed on their blood or other body fluids.

140. *(a)* Both A and R are true and R is the correct explanation of A.
The primary function of the Golgi apparatus is indeed to package proteins and lipids produced by the Endoplasmic Reticulum (ER) and deliver them to their target destinations, either within the cell or outside.
Vesicles containing materials from the ER fuse with the *cis* face of the Golgi apparatus. The contents are modified within the Golgi apparatus and are then transported and released from the *trans* face to various intracellular targets or secreted outside the cell.

141. *(a)* The correct match is A-II, B-III, C-IV, D-I.
The cotyledon of monocot seeds such as in grasses like wheat and maize is called scutellum.
Non-albuminous seed lacks endosperm, and the cotyledons store food, like in groundnut.
Epiblast is a rudimentary cotyledon, found in certain seeds.
Perisperm is the persistent nucellus, which functions as a storage tissue in some seeds.

142. *(a)* Both A and R are true and R is the correct explanation of A.
All vertebrates are chordates, but not all chordates are vertebrates. This is true, as vertebrates are a subgroup of chordates.
The members of subphylum-Vertebrata possess a notochord during the embryonic period, which is replaced by a cartilaginous or bony vertebral column in adults.

143. *(c)* Incorrect statement is given in option (c) and can be corrected as
The antigen-binding site of an antibody is located at the N-terminal region of the variable regions of both the heavy and light chains. The C-terminal region is part of the constant region of the antibody.

144. *(a)* RNA interference (RNAi) involves the silencing of specific *m*RNA through the use of complementary double-stranded RNA (dsRNA). The dsRNA is processed into small interfering RNAs (siRNAs) that guide the RNA-Induced Silencing Complex (RISC) to degrade or inhibit the target mRNA with complementary sequences, effectively silencing gene expression.

145. *(c)* Since genes R and Y follow independent assortment, the F_1 generation will be heterozygous for both traits (RrYy). In the F_2 generation, a dihybrid cross will occur, and the phenotypic ratio will follow the Mendelian pattern for two traits. The resulting phenotypic ratio for seed shape and colour will be 9 (round yellow) : 3 (round green) : 3 (wrinkled yellow) : 1 (wrinkled green).

146. *(a)* Histones are proteins that play a key role in the packaging of DNA in the nucleus. They are rich in basic amino

acids like lysine and arginine. These amino acids are positively charged, which helps in the binding of histones to the negatively charged phosphate backbone of DNA, aiding in DNA packaging and regulation.

147. *(b)* Menarche is the term used to describe the first occurrence of menstruation in females. It marks the onset of the reproductive phase in a woman's life.

148. *(d)* The correct match is A-III, B-I, C-IV, D-II.

Atrial natriuretic factor is produced by the heart to regulate blood pressure.
Erythropoietin is produced by the kidneys to stimulate red blood cell production.
Secretin is produced in the gastrointestinal tract to regulate digestion.
Aldosterone is produced by the adrenal cortex to regulate sodium and water balance.

149. *(c)* The apoenzyme is the protein portion of an enzyme, which requires a non-protein component (such as a cofactor or coenzyme) to become fully active. When the apoenzyme combines with its cofactor or coenzyme, it forms the complete active enzyme, known as a holoenzyme.

150. *(d)* The productivity of an ecosystem is typically measured in terms of energy captured by producers (like plants) per unit area over a time period. The unit $(kcal\ m^{-2})yr^{-1}$ represents the amount of energy (in kilocalories) produced or captured per square meter of the ecosystem per year. This is commonly used in ecological studies to express primary productivity.

151. *(a)* Sweet potato and potato represent analogous structures because they perform similar functions (both are storage organs), but they evolved independently in different plant families. This is an example of convergent evolution, where unrelated species develop similar traits due to similar environmental pressures or ecological functions, not due to a shared ancestry.

152. *(a)* In the F_1-generation, a carrier female X^cX and a non-affected normal male X^cY produced an affected male child, indicating that the genetic disorder follows a sex-linked recessive inheritance pattern. In the consanguineous mating between the female X^cY and male X^cY, out of four children, one was a carrier, which means the probability of having a carrier child is 1/4.

Possible offspring in F^1-generation; $X^c\ X^c \rightarrow$ affected female; $X^c\ X \rightarrow$ carrier female; $XY \rightarrow$ unaffected male; $X^cY \rightarrow$ affected male

	X^c	X
X^c	X^cX^c	X^cX
Y	X^cY	XY

153. *(a)* Both A and R are true and R is the correct explanation of A.
Tapetal cells are highly active metabolically and often become multinucleate (having more than one nucleus) through nuclear division without cytokinesis. This enhances their ability to produce and transport nutrients and other materials essential for the development of pollen (microspores).

154. *(c)* In angiosperms, the megaspore mother cell undergoes one meiotic division to produce four haploid megaspores, of which typically only one survives. This functional megaspore then undergoes three rounds of mitotic divisions to form the 8-nucleated, 7-celled embryo sac (female gametophyte).

155. *(c)* A zygomorphic flower is one that can be divided into two equal halves in only one plane. Pea flowers (family Fabaceae) are zygomorphic due to their bilaterally symmetrical structure. On the other hand, flowers like *Petunia*, *Datura*, and Chilli are actinomorphic, meaning they are radially symmetrical.

156. *(d)* The primary lymphoid organs are bone marrow and thymus where immature lymphocytes develop into antigen-sensitive cells. After maturation, these lymphocytes migrate to secondary lymphoid organs such as the spleen, lymph nodes, Peyer's patches of the small intestine, and appendix. These secondary organs serve as sites where lymphocytes interact with antigens to initiate immune responses.

157. *(b)* Both statement I and statement II are incorrect. They can be corrected as

Fig fruits may contain fig wasps that complete part of their life cycle inside the fruit, this does not make the fig "non-vegetarian." The wasps are part of a mutualistic relationship. The wasps typically die inside the fruit after completing their life cycle. In the process, it also pollinates the fig inflorescene (not fruit) of the fig. This ensures reproduction for both species—pollination for the fig and a habitat for the wasp's offspring.

158. *(a)* Spindle fibers are attached to the centromeres of chromosomes and are responsible for pulling the sister chromatids apart to opposite poles of the cell, ensuring that each daughter cell receives an identical set of chromosomes during cell division.

159. *(b)* In gymnosperms, seeds are not enclosed within a fruit (unlike angiosperms). Instead, they are exposed or "naked" on cone scales or other reproductive structures. This distinguishes them from angiosperms, where seeds are enclosed in a fruit.

160. *(b)* B, C and D are correct regarding function of adrenal medullary hormones. Incorrect option can be corrected as

Adrenal medullary hormones like adrenaline and noradrenaline do not cause pupillary constriction; they lead to pupillary dilation by stimulating the sympathetic nervous system, whereas constriction of the pupil is a function of the parasympathetic nervous system.

161. *(b)* Insulin is a protein hormone. If taken orally, it gets broken down by digestive enzymes in the stomach and intestines, just like other dietary proteins. This prevents it from entering the bloodstream effectively, making oral administration ineffective for managing blood glucose levels.

162. *(b)* The correct match is A-IV, B-III, C-I, D-II.

Salvinia is an aquatic fern, a type of pteridophyte.
Polytrichum is a moss, which is a bryophyte.
Salvia is a flowering plant, hence an angiosperm.
Ginkgo is a gymnosperm with naked seeds.

163. *(a)* George Gamow, was the first to propose that the genetic code is made up of triplets of nucleotides (three nucleotides per codon) to code for amino acids. He suggested this idea based on mathematical reasoning to explain how 4 nucleotides could code for 20 amino acids.

164. *(b)* The correct match is A-III, B-I, C-II, D-IV.

The Evil Quartet includes habitat loss, over-exploitation, alien species invasion, and co-extinctions. All these are major causes of biodiversity loss.
Ex situ conservation includes conservation strategies outside the natural habitat e.g., gene banks and cryopreservation techniques
Lantana camara is an invasive alien species causing harm to native biodiversity.
The *Dodo* is a well-known example of a species that has gone extinct.

165. *(b)* ADH also called vasopressin, is synthesised in the hypothalamus, specifically in the supraoptic and paraventricular nuclei, and then transported to and released from the posterior pituitary gland. Unlike other pituitary hormones such as LH, FSH, and ACTH, ADH is not synthesised in the pituitary itself.

166. *(b)* The water vascular system in echinoderms plays a crucial role in **respiration** (A) and the capture and transport of food (C). It helps with gas exchange (respiration) and uses the tube feet to aid in moving food towards the mouth, especially in species like sea stars. It does not directly assist with excretion, digestion, or locomotion in the way other systems might.

167. *(b)* The type of immunity present at the time of birth a non-specific defense mechanism is known as **innate immunity**. This type of immunity includes physical barriers like skin, as well as processes like inflammation and phagocytosis that help protect the body against pathogens from the time of birth.

168. *(b)* In bryophytes, gemmae are involved in asexual reproduction. Gemmae are small, multicellular structures that can develop into new organisms without the need for fertilisation.

169. *(d)* The renal portal system in frogs is a specialised venous connection that acts to link kidney and lower body parts. It collects deoxygenated blood from the hind limbs and the lower body. This blood is then directed to the kidneys for filtration before returning to the general circulation. This system is crucial for efficient waste removal from these body parts.

170. *(c)* Statement I is correct but Statement II is incorrect. Incorrect statement can be corrected as

According to the second law, energy is always lost as heat in every energy transfer, and this loss reduces the amount of useful energy available in the system. Even though ecosystems appear to have a constant flow of energy, they still follow the laws of thermodynamics, including the inevitable loss of energy as heat.

171. *(d)* RuBisCO is an enzyme that plays a key role in the Calvin cycle of photosynthesis. It helps in fixing carbon dioxide by catalysing the reaction between carbon dioxide and RuBP. This reaction forms two molecules of 3-phosphoglycerate, which are used to produce glucose and other organic compounds in plants. RuBisCo is a light activated enzyme and shows higher affinity for CO_2.

172. *(a)* For gene cloning, DNA mutase and DNA recombinase are not essential, while restriction enzymes are crucial for cutting DNA at specific sites to obtain the desired gene or fragment. DNA ligase is necessary for joining the DNA fragments together, especially when inserting the gene into a vector. DNA polymerase is important for amplifying DNA during PCR (Polymerase Chain Reaction), which is often used in gene cloning.

173. *(a)* Statements A, B and C are correct regarding plant growth and development. Statements D and E are incorrect and these can be corrected as
Abscisic acid is a growth inhibitor, often involved in stress responses, promoting processes like seed dormancy and inhibiting growth.
Apical dominance refers to the inhibition of lateral bud growth by the apical (top) bud. If the apical bud is removed, lateral buds can grow more freely.

174. *(c)* ρ (rho) is a protein that is important for the termination of transcription in prokaryotes. It binds to the RNA and moves along the transcript, eventually causing the RNA polymerase to dissociate from the DNA template, thereby terminating transcription.

175. *(c)* In water,frogs primarily respire through their skin. The lungs and buccal cavity are not very effective in water because the skin is the primary site for gas exchange.
On land, Frogs use their skin, buccal cavity and lungs for respiration. Lungs become more important on land because cutaneous respiration alone is not sufficient for the oxygen demands.
So, the statement is false for water, but true for land.

176. *(b)* Fraternal twins (dizygotic twins) are formed when two separate eggs are fertilised by two different sperm cells. Fraternal twins can be of the same sex or different sexes, making it possible for one to be a boy and the other a girl.

177. *(b)* *Aspergillus niger* is industrially vital for citric acid production, not household goods.
Lactobacillus is key bacterium for curd formation.
Trichoderma polysporum yields the immunosuppressant Cyclosporin-A, not household products.
Saccharomyces cerevisiae is used in baking and alcohol fermentation.
Propionibacterium shermanii is essential for Swiss cheese production.
Thus, *Aspergillus niger* and *Trichoderma polysporum* are primarily for industrial/pharmaceutical uses, not household product preparation.

178. *(a)*The correct match is A-IV, B-II, C-I, D-III.
Progesterone is a steroid hormone which is secreted by the corpus luteum. Relaxin is a protein hormone that is produced by the ovaries, particularly in the later stages of pregnancy. Melanocyte-stimulating hormone (MSH) is a Protein Hormone that is released by the pars intermedia. Catecholamines are amino-acid derived hormones which are released by the adrenal medulla during the times of stress.

179. *(d)* Statement I is incorrect but statement II is correct. Incorrect statement can be corrected as
Blue-coloured colonies indicate that the *Lac* Z gene is intact, meaning the plasmid does not contain a DNA insert. These are non-recombinant colonies because the insertion of foreign DNA disrupts the *Lac* Z gene, preventing the blue colour from forming.

180. *(b)* The Verhulst-Pearl Logistic Growth model describes how a population grows more rapidly, when the population size is small, but slows down as it approaches the carrying capacity (K) of the environment.
So the equation is:

$$\frac{dN}{di} = rN\left(\frac{K-N}{K}\right)$$

Where, N is the population size at time t; r is the intrinsic growth rate of the population; K is the carrying capacity.